

Power Management and Control of an Islanded Hybrid Power System

Matlab Simscape Modeling of Solar Panel, Fuel Cell and Battery with DC-DC Converter

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Abstract

In this thesis project, the development of a power management and control system for an islanded hybrid power system integrating a Photovoltaic (PV) panel, a fuel cell and a battery is explored. The project utilizes MATLAB Simulink, specifically the Simscape package, to model the hybrid power system. The main objectives are to enhance solar power utilization, ensure reliable power availability, and reduce residential electricity costs while maintaining the operational constraints of the battery and fuel cells.

Key components of the system include a Solar PV implementing Maximum Power Point Tracking (MPPT) to maximize solar power efficiency, simplified models of the fuel cell and battery to study interactions within the hybrid system, and DC-DC converter models utilizing boost and bidirectional buck-boost converters to manage energy flow and voltage regulation.

The power management system (PMS) utilizes both state flow and linear programming methods to balance power generation and consumption, ensuring robust and safe operation. The control system integrates flatness control and PID control to maintain the DC bus voltage within desired limits.

This project contributes to the ongoing development of hybrid renewable energy systems and smart grid technologies, offering more efficient and reliable power management solutions for residential applications.

Keywords: Hybrid System, Simscape, Photovoltaics, Fuel Cell, Battery, DC-DC Converter, MPPT, Stateflow, Linear Programming, Flatness, PID.

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Nomenclature

DC Direct Current

DES Discrete Event Systems

FC Fuel Cells

FPGA Field Programmable Gate Arrays

IC Incremental Conductance

IGBT Insulated Gate Bipolar Transistor

LP Linear Programming

MPPT Maximum Power Point Tracking

PID Proportional Integral Differential

PO Perturbed and Observe

PV Photovoltaics

PWM Pulse Width Modulation

PMS Power Management System

RE Renewable Energy

SOC State of Charge

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Chapter 1 - Introduction

1.1. Background

In recent years, there has been a significant shift towards obtaining energy for residential use with various alternative sources. Traditionally, household electricity is acquired through grid networks supplied by utility operators. However, Renewable Energy (RE) including solar photovoltaic (PV) systems, have emerged as greener options. The increasing interest in RE is driven by its sustainability, energy security, and environmental advantages, making it a crucial component of modern energy strategies (Renewable Energy Agency, 2020).

The main advantage of a power system with PV is its ability to provide a clean and renewable source of electricity at no cost during daylight hours. However, due to the discontinuity of power generation during nights or cloudy days, PV systems are often combined with fuel cells and batteries (Tesfahunegn, 2012). Batteries play a crucial role in balancing supply and demand by storing surplus energy and discharging it when needed, while fuel cells offer a reliable backup power source when both solar and battery energy are insufficient.

Although PV energy is free, several issues need to be addressed, particularly regarding capital and maintenance costs. To manage these concerns, it is important to develop strategies that ensure the system operates safely and extends the equipment's lifespan. Specifically, it is crucial to keep the battery within its State of Charge (SOC) range and to avoid exceeding the dynamic limits of the fuel cell by controlling the increase of its current. These measures help maintain both system's performance and durability.

1.2. Project Methodology

For simplicity, this master project will focus on developing power management and control for an islanded system containing PV, fuel cells and battery systems as shown in figure 1. In this project, modelling and simulations will be conducted using MATLAB and Simulink software (MathWorks, 2023).

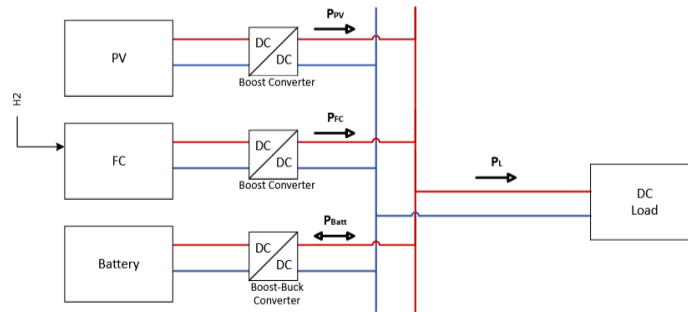


Figure 1. Islanded Hybrid Power System

The package used in Simulink is called Simscape which is graphical-based electrical model. Simscape is chosen because it provides high accuracy for modeling electrical systems and

has been successfully employed in similar studies focusing on renewable energy integration. Simscape Electrical offers comprehensive libraries for modeling and simulating power electronic, making it a powerful tool for developing and testing the power management and control systems (MathWorks, 2023a).

In this project, the control regulation of DC bus will be evaluated which requires a detailed analysis of the system's behavior. Since the main objective of this project is to focus on management and control, Matlab Simscape is effective in providing a comprehensive dynamic model of this hybrid power system. However, the high accuracy of Simscape comes with a drawback, as the high sample rate needed which can slow down the simulation process. This represent a disadvantage compared to using mathematical models, which can run faster by employing reduced-order or averages models.

While detailed mathematical model of the equipment will not be investigated, since Simscape is used, the mathematical model of the converter still needs to be evaluated, as it will be fundamental for deriving the control equations (Chapter 2).

1.3. Objectives and Project Questions

The primary objective of this power management and control system is to increase the utilization of solar power and the availability of power through the battery, thereby reducing residential electricity costs while respecting the operational constraints of both the battery and fuel cells.

The project focuses on two main areas:

Q1. Developing an effective lower-level control strategy for hybrid power systems

Q1.1. What control methods that can stabilize the voltage of the DC bus with good quality output?

Q1.2. How do different control strategies compare in terms of performance and efficiency? What solutions can be proposed based on this comparison?

Q2. Designing an efficient power management strategy that ensures the system operates within desired constraints and conditions

Q2.1. What power management systems can provide robust and safe operation, with potential for optimization?

Q2.2. What are the advantages and disadvantages of heuristic versus non-heuristic power management systems in the context of hybrid renewable energy systems?

By addressing these questions, this project aims to contribute to ongoing research in hybrid RE integration and smart grid technologies, potentially offering insights into more efficient and reliable hybrid power systems for residential applications.

Chapter 2 - Modeling and Proposed Control

This chapter explores the modeling and control strategies for the hybrid power system using Simscape. As illustrated in Figure 2, the power system model designed in Simscape integrates power sources, converters and loads. This modeling aims to accurately represent the hybrid system's performance under different conditions.

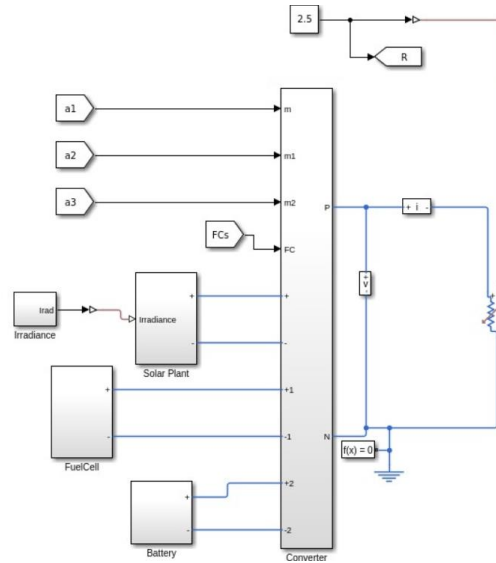


Figure 2. Simscape Model of the Hybrid Power System in Matlab Simulink

The chapter also examined the proposed control strategies for duty cycles inputs for the converter (a1, a2 and a3) and a contact switch for the fuel cells. It is important to note that the contact switches for the solar panel and the battery are not included in the model, as they are assumed to be always turned on.

2.1. Power Source Model

2.1.a. Model of Solar Panel

In general, the solar panels are typically characterized by their Power-Voltage (P-V) or Current-Voltage (I-V) curves. For this project, a solar PV panel with specific P-V characteristics is utilized. The P-V characteristics of the solar panel used in this project are illustrated in Figure 3.

The curve illustrates how the power output of the solar panel varies with voltage at the different irradiance levels. The table accompanying the curve provide the exact values of the maximum power point for that each irradiance levels. To simplify the model, temperature variance is not included, and is assumed to be constant.

Irradiance (W/m ²)	100	200	300	400	500	600	700	800	900	1000
Max Power (W)	113.4	240.8	373.5	510.4	649.8	790.9	933.9	1079	1224	1371

Table 1. Maximum Power at Ten Difference Irradiance Levels

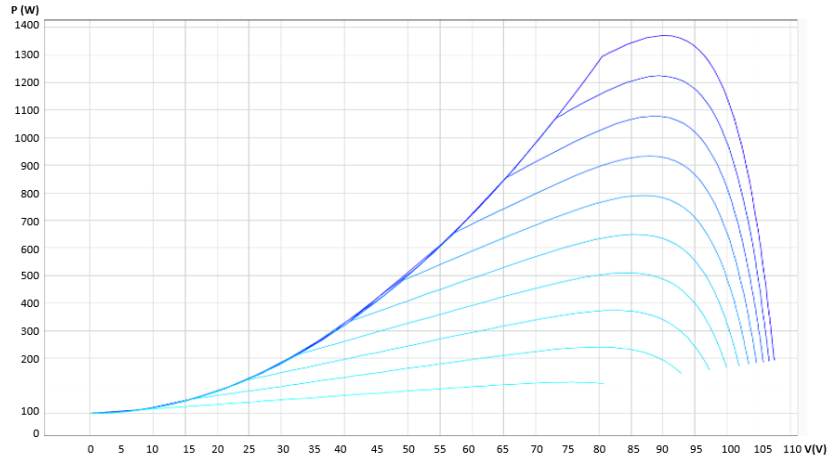


Figure 3. P-V Characteristics of the Solar Panel

Solar power is prioritized as the main power source due to its abundance and zero operational cost. Maximizing solar power utilization is essential, and therefore, a Maximum Power Point Tracking (MPPT) algorithm should be implemented. MPPT is crucial for optimizing the efficiency of PV systems and is a standard practice in PV power systems. This algorithm aims to locate the maximum power point under varying conditions due to irradiance or temperature changes. Some commonly used MPPT techniques include:

- Perturb and Observe (PO): This method perturbs the operating point of the PV system and observes the resulting change in power output. However, P&O can experience oscillations around the maximum power point, especially under rapidly changing environmental conditions (Hohm & Ropp, 2003)
- Incremental Conductance (IC): This method analyses the relationship between incremental changes in current and voltage to determine the maximum power point. It is generally more accurate than P&O, particularly in varying conditions, but it has more computational cost. Cao et al. (2017) has shown that IC can effectively minimize oscillations and improving tracking efficiency.

This project utilizes the IC MPPT algorithm, with the code provided in the appendix.

2.1.b. Model of Fuel Cell and Battery

In this project, idealized models of a fuel cell and a battery are implemented in MATLAB/Simscape, as illustrated in Figure 4. These models provide a simplified yet effective representation of the power sources behavior within the hybrid system.

The fuel cell model, shown on the left, is designed to simulate the electrochemical process of converting hydrogen into electrical energy. In this idealized representation, the essential voltage-current characteristics of a fuel stack allow for the study of its interaction with other power system components.

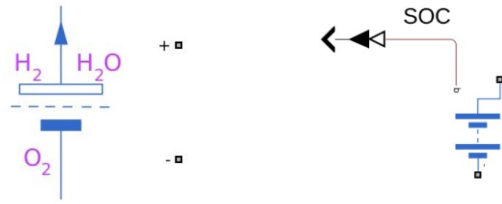


Figure 4. Simscape Models of the Fuel Cell (Above, and Left) and the Battery (Right)

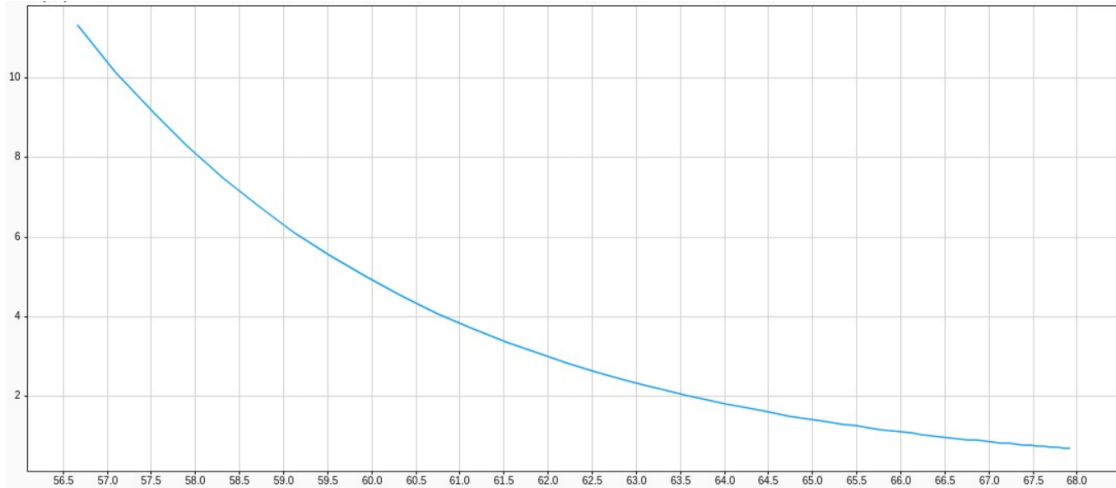


Figure 5. I-V Characteristics of the Fuel Cell

The battery model, depicted on the right, incorporates a state of charge (SOC) component, which is crucial for accurately representing the battery's energy storage capacity and its dynamic behavior during charging and discharging cycles.

Parameter	Fuel Cell	Battery
Nominal Voltage	50V	50 V
Rated Power	4000W	-
Capacity	-	11.6 Hr.A
Initial SOC	-	8 Hr.A
Internal Resistance	0.001 Ω	0.01 Ω

Table 2. Power Sources Characteristics

These idealized models, while not representing all the complexities of a real fuel cell and battery, provide a suitable framework for developing and testing power management and control strategies.

2.2. Converter Model

The hybrid system utilizes a combination of power electronic converters to manage the energy flow between various sources and the load. The converter configuration consists of two boost converters and one bidirectional buck-boost converter, as illustrated in Figure 5.

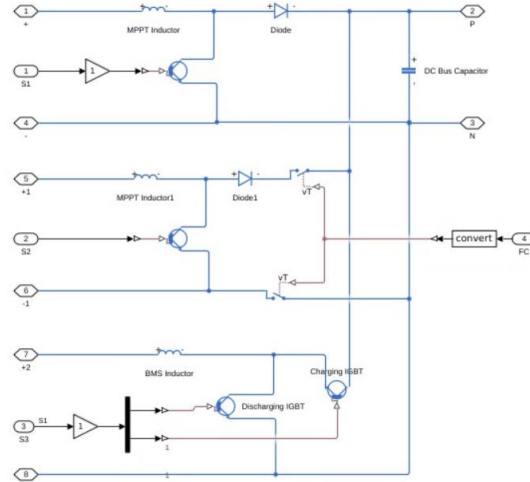


Figure 6. Converters Scheme in Matlab Simulink

The converter topology is designed to optimize power flow and voltage regulation within the hybrid system:

1. **Boost Converters:** Two boost converters are employed for the PV array and the fuel cell respectively. These converters step up the voltage from the sources to match the DC bus voltage, ensuring efficient power transfer and maximum power point tracking for the PV (Erickson & Maksimovic, 2001).
2. **Bidirectional Buck-Boost Converter:** This converter is connected to the battery, enabling bidirectional current flow for both charging and discharging operations. The buck-boost topology allows the battery voltage to be either stepped up or down as required, providing flexibility in battery operation (Hart, 2011)

The converters are modelled using nearly idealized components to focus on the system-level behavior and control strategies. The IGBT switches and diodes are represented with very low on-state resistances to approximate ideal switching behavior. The key parameters and components of the converters are summarized in Table below:

Parameter	PV Converter (Boost)	Fuel Cell Converter (Boost)	Battery Converter (Buck-Boost)
Inductance	0.0797 H	0.2677 H	0.004 H
Switching Frequency	20 kHz	20 kHz	20 kHz
IGBT On-Resistance	1e-6 Ω	1e-12 Ω	1e-6 Ω
Diode Forward Voltage	0.1 V	0.1 V	0.1 V
Bus Capacitance	0.015 F		

Table 3. Characteristics of Converter Components

This converter configuration allows for efficient power management and control voltage regulation within the hybrid system. The boost converters ensure that the output from the PV and fuel cells are compatible with the DC bus voltage, while the bidirectional buck-boost converter enables flexible energy storage and retrieval from the battery. The idealized component models facilitate the study of power management and control without addressing the detailed complexities of each component.

2.3. Power Management and Control Systems

The power management system is a critical component in hybrid energy systems, optimizing the generation, distribution, and utilization of energy from various sources such as solar panels, wind turbines and fuel cells. In general, it can employ various methods, ranging from simple rule-based (heuristic) approaches to complex optimization algorithms (non-heuristic methods), depending on the system's complexity and specific requirements.

- **Heuristic Methods**

Heuristic methods are based on experience, rules of thumb, and intuitive judgement. They are typically simpler to implement and can be very effective in practice, though they may not always provide the optimal solution. These methods rely on predefined rules or algorithms to make decisions about power management.

Examples of heuristic methods in power management include rule-based control and fuzzy logic control. Baumann et al. (2010) use a rule-base strategy to control a hybrid system with renewable energy sources, including micro wind turbines, fuel cells, PV and electrolyzers, utilizing flowchart and state charts. Another example is Karami et al. (2014), which discusses about energy management for PEMFC-PV system connected to a battery, super capacitor and the grid using a flowchart to manage the hybrid power system.

- **Non-heuristic methods**

This method relies on mathematical models and algorithms to find the optimal solution for power management, and are therefore also known as optimization-based methods. These methods are typically more complex and computationally intensive but can provide more accurate and efficient solutions. Examples of non-heuristic methods include linear/nonlinear programming and model predictive control (MPC). K/Bidi et al. (2022) design a MPS with MIQP optimization, using distributed explicit MPC (DeMPC) for voltage regulation.

In this project implements a two-level system architecture

Higher Level: Evaluates the power output from the solar panels (P_{pv}) and the fuel cell (P_{fc}).
Lower Level: Evaluates the battery's power output (P_{batt}) to stabilize the DC bus voltage.

The higher level plays a critical role in assessing the power contributions from both the solar panel and the fuel cell, while the inner level focuses on managing the battery's output to ensure a stable DC bus voltage.

2.3.a. Higher level Strategy: Power Management of Solar Panel and Fuel Cell

The primary function of this loop is to balance power generation with consumption, maximizing or minimizing the use of renewable energy. In this project, this level includes the management of power for solar panel and the fuel cell.

Further discussion on the choice of this loop strategy will be presented in Chapter 4. The selection of methods depends on factors such as system complexity, computational resources and specific application requirements.

Simplified Strategy

With the implementation of MPPT, solar power (P_{pv}) is treated as an independent variable or a disturbance, as is the power load (P_L). The PMS will continuously implement MPPT unless the battery's SOC reaches its maximum capacity. In Chapter 4 where methods of management system will be discussed more comprehensively.

The fundamental power balance equation for the DC bus is

$$P_{Sources} = P_{Demand} \quad (1)$$

$$P_{bus} + P_{FC} + P_{Batt} = P_L - P_{PV} \quad (2)$$

The primary objective of the PMS is to ensure that the left side of equation (2) supplies sufficient power when $P_{demand} > 0$ or absorbs excess power when $P_{demand} < 0$.

In the next section, the development of the lower level strategy will be discussed and tested. Therefore, a temporary scheme of this higher level strategy is necessary. For the initial analysis, a simplified rule is defined for the following condition:

$$P_{FC} = < P_{Load} - P_{PV} > \quad (3)$$

In this project, the fuel cell's dynamic response is limited to preserve its lifetime by implementing a current input filter. The $< >$ notation indicates that the power of the fuel cell is the rated power difference between the load power and the solar power. This means that the power of the fuel cell is the output after $P_{Load} - P_{PV}$ is filtered. A Butterworth filter is used for this purpose.

It is important to note that by stating condition (3), a basic rule-based method has already been implemented, as the fuel cell power is defined by this simply rule. This simplified approach allows us to focus on evaluation the DC bus voltage in the subsequent sections. More developed of outer loop strategies will be explored in Chapter 4.

2.3.b. Lower Level Strategy: Power Management of Battery

The main objective of this strategy is to stabilize the DC bus voltage (V_{dc}) around its reference with an error of approximately 5%. Controlling V_{dc} is crucial because, during transient conditions, equation (2) has behavior as (4).

$$P_{bus} \neq 0 \quad (4)$$

Flatness Control

The concept of differential flatness has become a powerful tool in nonlinear control theory. A system is considered differentially flat if there exists a set of variables, called flat outputs, equal in number to the system inputs, such that all system states and inputs can be expressed as functions of these flat outputs and their derivatives. This property allows for easier trajectory planning and feedback linearization of nonlinear systems, making it particularly useful for control design (Fliess et al., 1994).

To address problem (4), Thounthong et al (2014) has successfully provide a solution using flatness equations. Suppose that there is a flat output y , control input u , and state variable x defined as

$$y = E_{bus}, \quad u = P_{Batt}^*, \quad x = V_{DC} \quad (5)$$

From the capacitance equation, the relationship between E_{bus} and V_{dc} is given by:

$$E_{Bus} = \frac{1}{2} C V_{DC}^2 \quad (6)$$

Thus, the state variable x can be expressed as:

$$x = \sqrt{\frac{2y}{C_{bus}}} = \varphi(y) \quad (7)$$

From the power balance equation:

$$P_{Batt} = \dot{E} - P_{PV} - P_{FC} + P_{Load} \quad (8)$$

This yields the flat output:

$$u = P_{Batt}^* = \dot{y} - P_{PV} - P_{FC} + \sqrt{\frac{2y}{C_{bus}}} i_{Load} = \psi(y, \dot{y}) \quad (9)$$

Since $x = \varphi(y)$ and there is function $u = \psi(y, \dot{y})$, it is clear that y is a flat output. Finally, using the feedback linearization formula, exponential asymptotic tracking of the trajectory y^* can be achieved:

$$(\dot{y} - \dot{y}^*) + Kp(y - y^*) + Ki \int (y - y^*) d\tau = 0 \quad (10)$$

By substituting $\dot{y}$ from (10) back into (9) the input for battery P_{Batt}^* can be calculated .

2.4. Lower Level Strategy: Flatness-PID

In this section, the integration of the simplified outer strategy discussed earlier and the inner loop system for control and stabilize DC bus voltage control will be summarized.

In Thounthong et al. (2014), the method for defining the input for the converter, specifically the duty cycle using the reference P_{Batt}^* , is not clearly explained. To address this, we propose implementing a PID control strategy to manage the duty cycle for both the fuel cell and the battery. Additionally, it is crucial for the control system to determine when the battery should operate in buck mode and when it should switch to boost mode. To achieve this, a selector mechanism will be integrated into the control system.

In summary, by integrating simplified power management (3) and control system equation (9) the overall scheme will be as follows:

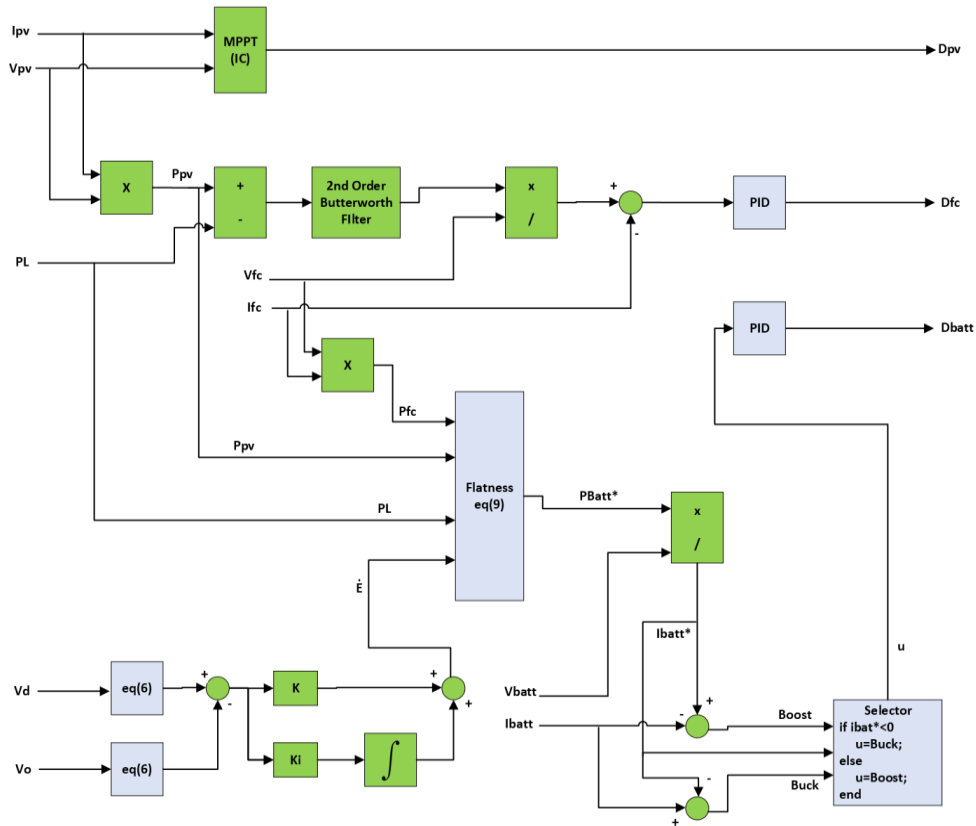


Figure 7. Power Management and Control of Hybrid System using Flatness and PID

From the figure above, it can be seen that the complete scheme for the outer loop of the PMS consist of the MPPT, the rated power for fuel cell, and the flatness (9) with its feedback linearization. The inner loop includes the PIDs and a selector.

2.5. Lower Level Strategy Strategy: Double Flatness

This subsection explores a modified flatness control strategy based on the average model converters, derived from the work of Messaoudi et al. (2024). The approach, termed “double flatness”, involves significant modifications to adapt the control system to the specific hybrid power system configuration and performance requirements.

In the referenced paper, a configuration of two boost converters and one buck-boost converter is analysed, with the buck-boost converter positioned as the 2nd state. However, in this part, the buck-boost converter (representing the battery) has been repositioned as the third state. This modification implies changes to the average model and affects the flatness output formulation. The repositioning is important because the trajectory assigned to the third state differs from that of the second state. In the referenced paper, the 3rd state trajectory is defined as:

$$P_3^* = P_{bus} - P1 - P2 \quad (15)$$

Where $\dot{P}_{bus} = \int -\lambda(V_{DC} - V_{DC}^*)$ and 1, 2, 3 represent any type of power sources.

However, this trajectory is unsuitable for this hybrid system due to its slow control response. Instead, the trajectory defined in equation (9) is preferred, as it offers improved dynamic performance. This modification in both the battery state positioning and trajectory formulation constitutes a key contribution of this project.

After these modifications, the average model of the converters configured as illustrated in Figure 5 is represented as bellow.

Boost:

$$\begin{bmatrix} \dot{I}_{pv} \\ \dot{I}_{fc} \\ \dot{I}_{batt} \\ \dot{V}_{DC} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \frac{\alpha_{pv}-1}{L_{pv}} \\ 0 & 0 & 0 & \frac{\alpha_{fc}-1}{L_{fc}} \\ 0 & 0 & 0 & \frac{\alpha_{batt}-1}{L_{batt}} \\ \frac{1-\alpha_{pv}}{C} & \frac{1-\alpha_{fc}}{C} & \frac{1-\alpha_{batt}}{C} & \frac{-1}{RC} \end{bmatrix} \begin{bmatrix} I_{pv} \\ I_{fc} \\ I_3 \\ V_{DC} \end{bmatrix} + \begin{bmatrix} \frac{V_{pv}}{L_{pv}} \\ \frac{V_{fc}}{L_{fc}} \\ \frac{V_{batt}}{L_{batt}} \\ 0 \end{bmatrix} \quad (11)$$

Buck:

$$\begin{bmatrix} \dot{I}_{pv} \\ \dot{I}_{fc} \\ \dot{I}_{batt} \\ \dot{V}_{DC} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \frac{\alpha_{pv}-1}{L_{pv}} \\ 0 & 0 & 0 & \frac{\alpha_{fc}-1}{L_{fc}} \\ 0 & 0 & 0 & -\frac{\alpha_{batt}}{L_{batt}} \\ \frac{1-\alpha_{pv}}{C} & \frac{1-\alpha_{fc}}{C} & \frac{\alpha_{batt}}{C} & \frac{-1}{RC} \end{bmatrix} \begin{bmatrix} I_{pv} \\ I_{fc} \\ I_3 \\ V_{DC} \end{bmatrix} + \begin{bmatrix} \frac{V_{pv}}{L_{pv}} \\ \frac{V_{fc}}{L_{fc}} \\ \frac{V_{batt}}{L_{batt}} \\ 0 \end{bmatrix} \quad (12)$$

Messaoudi et al. (2024) demonstrate that the system described by equations (11), (12) is uniformly flat around any $x^* \in \mathbb{R}_+^* \times \mathbb{R} \times \mathbb{R}_+^* \times \mathbb{R}_+^*$ and the three-tuple $(h_{pv}, h_{fc}, h_{batt})$ of smooths functions given by:

$$\begin{aligned} h_{pv}(x) &= x_{pv} = I_{pv} \\ h_{fc}(x) &= x_{fc} = I_{fc} \\ h_{batt}(x) &= \frac{1}{2}(L_{pv}I_{pv}^2 + L_{fc}I_{fc}^2 + L_{batt}I_{batt}^2 + CV_{DC}^2) \end{aligned}$$

Is defines a flat output of system (11), (12) at x^* .

Flatness Proof

To prove flatness, consider the time derivative of h_{batt} :

$$\dot{h}_{batt} = L_{pv}I_{pv}\dot{I}_{pv} + L_{fc}I_{fc}\dot{I}_{fc} + L_{batt}I_{batt}\dot{I}_{batt} + CV_{DC}\dot{V}_{DC}$$

Substituting the $\dot{I}_{pv}, \dot{I}_{fc}, \dot{I}_{batt}, \dot{V}_{DC}$ from the average model, we obtain:

• Boost Converter

$$\begin{aligned} \dot{h}_{batt} = & L_{pv}I_{pv}\left(\frac{\alpha_{pv}-1}{L_{pv}}V_{DC} + \frac{V_{pv}}{L_{pv}}\right) + L_{fc}I_{fc}\left(\frac{\alpha_{fc}-1}{L_{fc}}V_{DC} + \frac{V_{fc}}{L_{fc}}\right) + L_{batt}I_{batt}\left(\frac{\alpha_{batt}-1}{L_{batt}}V_{DC} + \frac{V_{batt}}{L_{batt}}\right) + \\ & CV_{DC}\left(\frac{1-\alpha_{pv}}{C}I_{pv} + \frac{1-\alpha_{fc}}{C}I_{fc} + \frac{1-\alpha_{batt}}{C}I_{batt} - \frac{V_{DC}}{RC}\right) \end{aligned}$$

which simplifies to:

$$\dot{h}_{batt} = V_{pv} \cdot I_{pv} + V_{fc} \cdot I_{fc} + V_{batt} \cdot I_{batt} - \frac{V_{DC}^2}{R}$$

By substituting back V_{DC}^2 to h_{batt} equation,

$$h_{batt} = \frac{1}{2}(L_{pv}I_{pv}^2 + L_{fc}I_{fc}^2 + L_{batt}I_{batt}^2 + RC(V_{pv} \cdot I_{pv} + V_{fc} \cdot I_{fc} + V_{batt} \cdot I_{batt} - \dot{h}_{batt}))$$

leading to:

$$L_{pv}I_{pv}^2 + L_{fc}I_{fc}^2 + L_{batt}I_{batt}^2 + RC(V_{pv} \cdot I_{pv} + V_{fc} \cdot I_{fc} + V_{batt} \cdot I_{batt} - \dot{h}_{batt}) - 2h_{batt} = 0$$

$$\frac{L_{batt}}{C}I_{batt}^2 + R \cdot V_{batt} \cdot I_{batt} + \left(\frac{L_{pv}}{C}h_{pv}^2 + \frac{L_{fc}}{C}h_{fc}^2 + R(V_{pv} \cdot h_{pv} + V_{fc} \cdot h_{fc}) - R\dot{h}_{batt} - \frac{2}{C}h_{batt}\right) = 0$$

Further simplifying:

$$\frac{L_{batt}}{C}I_{batt}^2 + R \cdot V_{batt} \cdot I_{batt} + \mu(h, \dot{h}) = 0$$

which is a second-order polynomial equation in I_{batt}

$$I_{batt} = \frac{-RCV_{batt} + \sqrt{(RCV_{batt})^2 - 4L_{batt}C\mu(h, \dot{h})}}{2L_{batt}}$$

Similarly, by substituting I_{batt} back into h_{batt} equation, the DC bus voltage V_{DC} is given by:

$$V_{DC} = \sqrt{\frac{2}{C}h_{batt} - \frac{L_{pv}}{C}h_{pv}^2 - \frac{L_{fc}}{C}h_{fc}^2 + \frac{L_{batt}}{C}I_{batt}^2}$$

Therefore, since $I_{batt} = \gamma(h, \dot{h})$ and $V_{DC} = \varphi(h, \dot{h})$, it is evident that the system is flat.

• Buck Converter

For the buck converter, the expression for $\dot{h}_{batt}$ as follows:

$$\begin{aligned} \dot{h}_{batt} = & L_{pv}I_{pv}\left(\frac{\alpha_{pv}-1}{L_{pv}}V_{DC} + \frac{V_{pv}}{L_{pv}}\right) + L_{fc}I_{fc}\left(\frac{\alpha_{fc}-1}{L_{fc}}V_{DC} + \frac{V_{fc}}{L_{fc}}\right) + L_{batt}I_{batt}\left(-\frac{\alpha_{batt}}{L_{batt}}V_{DC} + \frac{V_{batt}}{L_{batt}}\right) + \\ & CV_{DC}\left(\frac{1-\alpha_{pv}}{C}I_{pv} + \frac{1-\alpha_{fc}}{C}I_{fc} + \frac{\alpha_{batt}}{C}I_{batt} - \frac{V_{DC}}{RC}\right) \end{aligned}$$

Which lead to the same $\dot{h}_{batt}$ result as boost converter:

$$\dot{h}_{batt} = V_{pv} \cdot I_{pv} + V_{fc} \cdot I_{fc} + V_{batt} \cdot I_{batt} - \frac{V_{DC}^2}{R}$$

Thus, by repeating the same procedure, flatness can also be demonstrated for the buck condition.

Duty Cycle Derivation

To derive the duty cycle for each switch, the method applied in Chapter 2 is reiterated:

- Duty cycle for PV

Given that $h_{pv}(x) = I_{pv}$ we have $\dot{h}_{pv}(x) = \dot{I}_{pv}$ from the average model equation:

$$\dot{I}_{pv} = \frac{\alpha_{pv} - 1}{L_{pv}}V_{DC} + \frac{V_{pv}}{L_{pv}}$$

Which gives

$$\alpha_{pv} = \frac{\dot{I}_{pv} \cdot L_{pv} - V_{pv} + V_{DC}}{V_{DC}} = \frac{L_{pv}}{V_{DC}} \left(\frac{V_{DC} - V_{pv}}{L_{pv}} + \dot{h}_{pv} \right)$$

Using feedback linearization formula, the duty cycle is:

$$\alpha_{pv} = \frac{L_{pv}}{V_0} \left(\frac{V_0 - V_{pv}}{L_{pv}} - \lambda_{pv}(h_{pv} - h_{dpv}) + \dot{h}_{dpv} \right)$$

- Duty cycle for FC

From the same steps as for PV and noting that $h_{fc}(x) = I_{fc}$ the duty cycle for the fuel cell is derived as:

$$\alpha_{fc} = \frac{L_{fc}}{V_0} \left(\frac{V_0 - V_{fc}}{L_{fc}} - \lambda_{fc}(h_{fc} - h_{d_{fc}}) + \dot{h}_{d_{fc}} \right)$$

- Duty cycle for Batt

Boost Condition

Starting back from the expression:

$$\dot{h}_{batt} = V_{pv} \cdot I_{pv} + V_{fc} \cdot I_{fc} + V_{batt} \cdot I_{batt} - \frac{V_{DC}^2}{R}$$

And differentiating once more:

$$\ddot{h}_{batt} = \dot{V}_{pv} \cdot I_{pv} + V_{pv} \cdot \dot{I}_{pv} + \dot{V}_{fc} \cdot I_{fc} + V_{fc} \cdot \dot{I}_{fc} + \dot{V}_{batt} \cdot I_{batt} + V_{batt} \cdot \dot{I}_{batt} - \frac{2V_{DC} \cdot \dot{V}_{DC}}{R}$$

Assuming that source voltages are constant or change very slow, this simplifies to:

$$\ddot{h}_{batt} = V_{pv} \cdot \dot{I}_{pv} + V_{fc} \cdot \dot{I}_{fc} + V_{batt} \cdot \dot{I}_{batt} - \frac{2V_{DC} \cdot \dot{V}_{DC}}{R}$$

By substituting the $\dot{I}_{pv}$, $\dot{I}_{fc}$, $\dot{I}_{batt}$, $\dot{V}_{DC}$ from the average model, the expression for $\ddot{h}_{batt}$ as follows:

$$\begin{aligned} \ddot{h}_{batt} = & V_{pv} \left(\frac{\alpha_{pv} - 1}{L_{pv}} V_{DC} + \frac{V_{pv}}{L_{pv}} \right) + V_{fc} \left(\frac{\alpha_{fc} - 1}{L_{fc}} V_{DC} + \frac{V_{fc}}{L_{fc}} \right) + V_{batt} \left(\frac{\alpha_{batt} - 1}{L_{batt}} V_{DC} \right. \\ & \left. + \frac{V_{batt}}{L_{batt}} \right) - \frac{2V_{DC}}{R} \left(\frac{1 - \alpha_{pv}}{C} I_{pv} + \frac{1 - \alpha_{fc}}{C} I_{fc} + \frac{1 - \alpha_{batt}}{C} I_{batt} - \frac{V_{DC}}{RC} \right) \end{aligned}$$

The duty cycle for the battery in boost condition is:

$$\begin{aligned} \alpha_{batt} = & \left(\left(\frac{V_{batt} V_{dc}}{L_{batt}} + \frac{2V_o I_{batt}}{RC} \right)^{-1} \left(-\frac{V_{pv}(V_{pv} - V_o)}{L_{pv}} - \frac{V_{fc}(V_{fc} - V_{dc})}{L_{fc}} \right. \right. \\ & \left. \left. - \frac{V_{batt}(V_{batt} - V_{dc})}{L_{batt}} + \frac{2V_{dc}}{R} \left(\frac{I_{pv} + I_{fc} + I_{batt}}{C} - \frac{V_{dc}}{RC} \right) \right. \right. \\ & \left. \left. - \left(\frac{V_{pv} V_o}{L_{pv}} + \frac{2V_{dc} I_{pv}}{RC} \right) \alpha_{pv} - \left(\frac{V_{fc} V_{dc}}{L_{fc}} + \frac{2V_{dc} I_{fc}}{RC} \right) \alpha_{fc} - \ddot{h}_{batt} \right) \end{aligned}$$

Using feedback linearization, the duty cycle can be obtained as equation (13) bellow.

Buck Condition

From the flatness property, we know that:

$$\dot{h}_{batt} = V_{pv} \cdot I_{pv} + V_{fc} \cdot I_{fc} + V_{batt} \cdot I_{batt} - \frac{V_{DC}^2}{R}$$

Differentiating this equation once again yields:

$$\ddot{h}_{batt} = \dot{V}_{pv} \cdot I_{pv} + V_{pv} \cdot \dot{I}_{pv} + \dot{V}_{fc} \cdot I_{fc} + V_{fc} \cdot \dot{I}_{fc} + \dot{V}_{batt} \cdot I_{batt} + V_{batt} \cdot \dot{I}_{batt} - \frac{2V_{DC} \cdot \dot{V}_{DC}}{R}$$

Assuming that the source voltages V_{pv}, V_{fc}, V_{batt} are constant or change very slow, the above equation simplifies to:

$$\ddot{h}_{batt} = V_{pv} \cdot \dot{I}_{pv} + V_{fc} \cdot \dot{I}_{fc} + V_{batt} \cdot \dot{I}_{batt} - \frac{2V_{DC} \cdot \dot{V}_{DC}}{R}$$

By substituting the $\dot{I}_{pv}, \dot{I}_{fc}, \dot{I}_{batt}, \dot{V}_{DC}$ from the average model, we obtained:

$$\begin{aligned} \ddot{h}_{batt} = & V_{pv} \left(\frac{\alpha_{pv} - 1}{L_{pv}} V_{DC} + \frac{V_{pv}}{L_{pv}} \right) + V_{fc} \left(\frac{\alpha_{fc} - 1}{L_{fc}} V_{DC} + \frac{V_{fc}}{L_{fc}} \right) + V_{batt} \left(-\frac{\alpha_{batt}}{L_{batt}} V_{DC} \right. \\ & \left. + \frac{V_{batt}}{L_{batt}} \right) - \frac{2V_{DC}}{R} \left(\frac{1 - \alpha_{pv}}{C} I_{pv} + \frac{1 - \alpha_{fc}}{C} I_{fc} + \frac{\alpha_{batt}}{C} I_{batt} - \frac{V_{DC}}{RC} \right) \end{aligned}$$

The duty cycle for the battery in the buck condition is then given by:

$$\begin{aligned} \alpha_{batt} = & \left(\frac{V_{batt} V_{dc}}{L_{batt}} + \frac{2V_{dc} I_{batt}}{RC} \right)^{-1} \left(\frac{V_{pv}(V_{pv} - V_{dc})}{L_{pv}} + \frac{V_{fc}(V_{fc} - V_{dc})}{L_{fc}} \right. \\ & \left. + \left(\frac{V_{pv} V_{dc}}{L_{pv}} + \frac{2V_{dc} I_{pv}}{RC} \right) \alpha_{pv} + \left(\frac{V_{fc} V_{dc}}{L_{fc}} + \frac{2V_{dc} I_{fc}}{RC} \right) \alpha_{fc} + \ddot{h}_{batt} \right) \end{aligned}$$

Using feedback linearization formula, the duty cycle can be obtained as expressed in equation (14) below.

Boost Mode:

$$\alpha_{pv} = \frac{L_{pv}}{V_{dc}} \left(\frac{V_{dc} - V_{pv}}{L_{pv}} - \lambda_{pv}(h_{pv} - h_{dpv}) + \dot{h}_{dpv} \right)$$

$$\alpha_{fc} = \frac{L_{fc}}{V_{dc}} \left(\frac{V_{dc} - V_{fc}}{L_{fc}} - \lambda_{fc}(h_{fc} - h_{dfc}) + \dot{h}_{dfc} \right)$$

(13)

$$\begin{aligned} \alpha_{batt} = & \left(\frac{V_{batt} V_{dc}}{L_{batt}} + \frac{2V_{dc} I_{batt}}{RC} \right)^{-1} \left(-\frac{V_{pv}(V_{pv} - V_o)}{L_{pv}} - \frac{V_{fc}(V_{fc} - V_{dc})}{L_{fc}} - \frac{V_{batt}(V_{batt} - V_{dc})}{L_{batt}} \right. \\ & \left. + \frac{2V_{dc}}{R} \left(\frac{I_{pv} + I_{fc} + I_{batt}}{C} - \frac{V_{dc}}{RC} \right) - \left(\frac{V_{pv} V_o}{L_{pv}} + \frac{2V_{dc} I_{pv}}{RC} \right) \alpha_{pv} - \left(\frac{V_{fc} V_{dc}}{L_{fc}} + \frac{2V_{dc} I_{fc}}{RC} \right) \alpha_{fc} \right) \end{aligned}$$

$$- \lambda_{batt,0}(h_{batt}-h_{dbatt}) - \lambda_{batt,1}(\dot{h}_{batt}-\dot{h}_{dbatt}) + \ddot{h}_{dbatt})$$

Buck Mode

$$\begin{aligned} \alpha_{pv} &= \frac{L_{pv}}{V_{dc}} \left(\frac{V_{dc} - V_{pv}}{L_{pv}} - \lambda_{pv}(h_{pv}-h_{dpv}) + \dot{h}_{dpv} \right) \\ \alpha_{fc} &= \frac{L_{fc}}{V_{dc}} \left(\frac{V_{dc} - V_{fc}}{L_{fc}} - \lambda_{fc}(h_{fc}-h_{dfc}) + \dot{h}_{dfc} \right) \\ \alpha_{batt} &= \left(\frac{V_{batt}V_{dc}}{L_{batt}} + \frac{2V_{dc}I_{batt}}{RC} \right)^{-1} \left(\frac{V_{pv}(V_{pv} - V_{dc})}{L_{pv}} + \frac{V_{fc}(V_{fc} - V_{dc})}{L_{fc}} + \left(\frac{V_{pv}V_{dc}}{L_{pv}} + \frac{2V_{dc}I_{pv}}{RC} \right) \alpha_{pv} \right. \\ &\quad \left. + \left(\frac{V_{fc}V_{dc}}{L_{fc}} + \frac{2V_{dc}I_{fc}}{RC} \right) \alpha_{fc} + \lambda_{batt,0}(h_{batt}-h_{dbatt}) + \lambda_{batt,1}(\dot{h}_{batt}-\dot{h}_{dbatt}) - \ddot{h}_{dbatt} \right) \end{aligned} \quad (14)$$

To complete the control system, the PV current reference is defined as:

$$I_{dpv} = \frac{I_{PV} \cdot V_{PV}}{V_{mppt}} \quad (15)$$

Where V_{mppt} is the voltage at the maximum power point.

In summary, the proposed management and control system, incorporating the double flatness approach, is illustrated in the following figure 7.

However, the performance of this double flatness should be evaluated further to prove whether it works better than the conventional flatness-PID. Further simulation and experimental validation will be conducted in chapter 3 to verify these improvements of this control method.

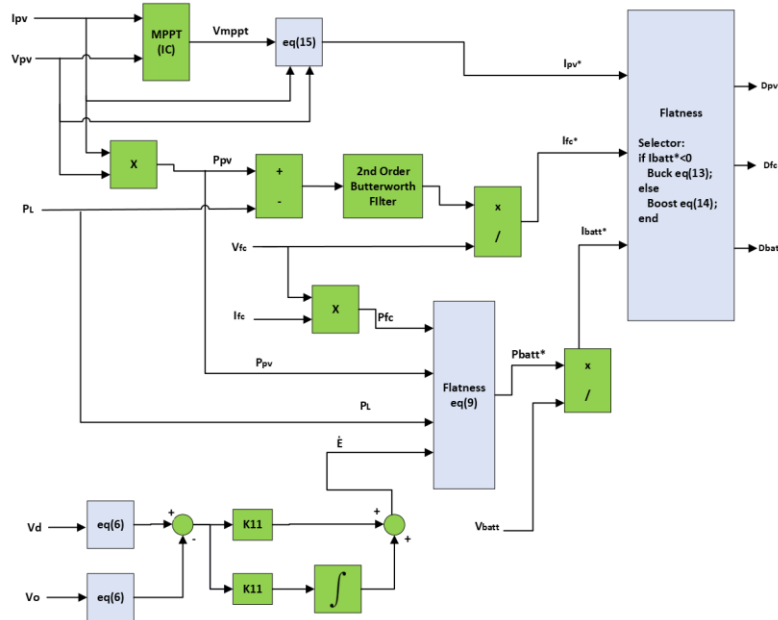


Figure 8. Power Management and Control of Hybrid System using Double Flatness

Chapter 3 – Control and Modeling Validation

3.1. MPPT Validation

MPPT validation is conducted by varying irradiance under stable load demand conditions. The simulation results compare the actual maximum power with the power obtained by the MPPT algorithm, as shown in Table 3.

Irradiance (W/m ²)	Max Power (W)	MPPT Power (W)
1000	1371	1371
900	1224	1224
800	1079	1078
700	933.9	933
600	790.9	790
500	649.8	649
400	510.4	510
300	373.5	373
200	240.8	240
100	113.4	113

Table 4. Comparison of Maximum Power and MPPT Power at Various Irradiance Levels

The results demonstrate that the system consistently operates at the maximum power point of the solar panel. The MPPT algorithm and the graphical representation of this validation are provided in the appendix.

3.2. Control Validation

This section presents the validation of the two control methods for lower level strategy discussed in chapter 2.4 and 2.5. The simulation aims to compare the performance between these two control strategies: Flatness-PID and Double flatness. Detailed transient response simulations are available in the appendix.

Figure 9 illustrates the comparison between the Flatness-PID and Double-Flatness control methods. The simulation results indicate that both methods exhibit similar performance characteristic. Both control strategies successfully stabilize the DC bus voltage despite significant disturbances exceeding 1000W

As observed in the graph, major disturbances in P_L occur at approximately 0.3s, 0.5s, 0.7s, 2s and 2.3s, while disturbances in P_{PV} occur at around 1.1s and 1.3s. The ability of both control systems to maintain stability under these conditions demonstrates their robustness and effectiveness in managing the hybrid power system.

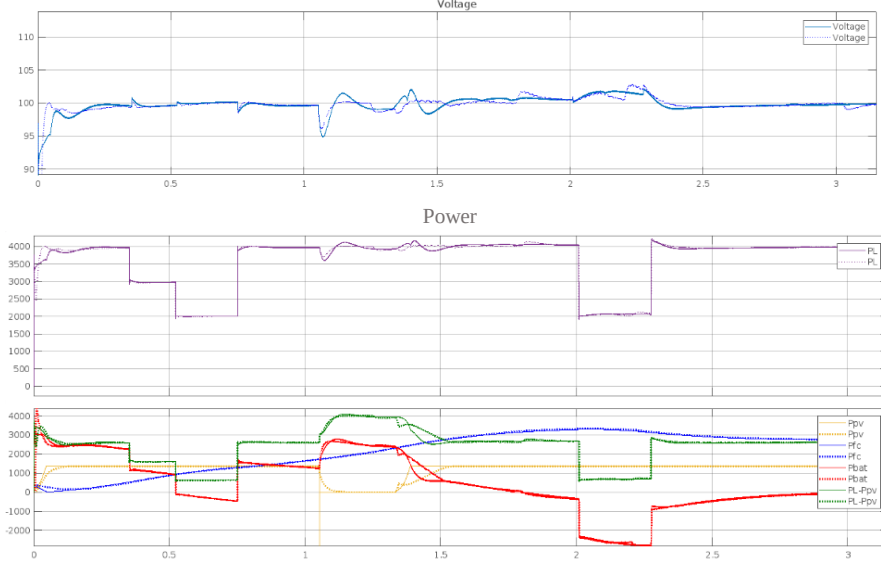


Figure 9. Comparison of Control with Flatness-PID (Dotted) and Double Flatness (Line)

3.3. Proposed Control

Inspired by the similar properties of the two previously discussed control methods, a new control strategy is proposed. This approach modifies the double flatness control discussed in chapter 2.5 so that the MPPT algorithm operates independently. The modifications to equations (13) and (14) are as follows:

Boost:

$$\alpha_{pv} = \alpha_{mppt}$$

$$\alpha_{fc} = \frac{L_{fc}}{V_0} \left(\frac{V_0 - V_{fc}}{L_{fc}} - \lambda_{fc}(h_{fc} - h_{afc}) + \dot{h}_{afc} \right) \quad (16)$$

$$\begin{aligned} \alpha_{batt} = & \left(\frac{V_{batt} V_{dc}}{L_{batt}} + \frac{2V_o I_{batt}}{RC} \right)^{-1} \left(-\frac{V_{pv}(V_{pv} - V_o)}{L_{pv}} - \frac{V_{fc}(V_{fc} - V_{dc})}{L_{fc}} - \frac{V_{batt}(V_{batt} - V_{dc})}{L_{batt}} \right. \\ & \left. + \frac{2V_{dc}}{R} \left(\frac{I_{pv} + I_{fc} + I_{batt}}{C} - \frac{V_{dc}}{RC} \right) - \left(\frac{V_{pv} V_o}{L_{pv}} + \frac{2V_{dc} I_{pv}}{RC} \right) \alpha_{pv} - \left(\frac{V_{fc} V_{dc}}{L_{fc}} + \frac{2V_{dc} I_{fc}}{RC} \right) \alpha_{fc} \right) \end{aligned}$$

Buck:

$$\alpha_{pv} = \alpha_{mppt} \quad (17)$$

$$\alpha_{fc} = \frac{L_{fc}}{V_0} \left(\frac{V_0 - V_{fc}}{L_{fc}} - \lambda_{fc}(h_{fc} - h_{afc}) + \dot{h}_{afc} \right)$$

$$\alpha_{batt} = \left(\frac{V_{batt} V_{dc}}{L_{batt}} + \frac{2V_{dc} I_{batt}}{RC} \right)^{-1} \left(\frac{V_{pv}(V_{pv} - V_{dc})}{L_{pv}} + \frac{V_{fc}(V_{fc} - V_{dc})}{L_{fc}} \right) + \left(\frac{V_{pv} V_{dc}}{L_{pv}} + \frac{2V_{dc} I_{pv}}{RC} \right) \alpha_{pv} + \left(\frac{V_{fc} V_{dc}}{L_{fc}} + \frac{2V_{dc} I_{fc}}{RC} \right) \alpha_{fc} + \lambda_{batt,0}(h_{batt} - h_{battn}) + \lambda_{batt,1}(\dot{h}_{batt} - \dot{h}_{dbatt}) - \ddot{h}_{dbatt}$$

Where $h_n = \frac{1}{2}(L_{fc} i_{fc}^2 + L_{batt} i_{batt}^2 + CV_o^2)$ because $I_{dpv} = I_{pv}$

This modification results in a new power management scheme, as illustrated in the following figure:

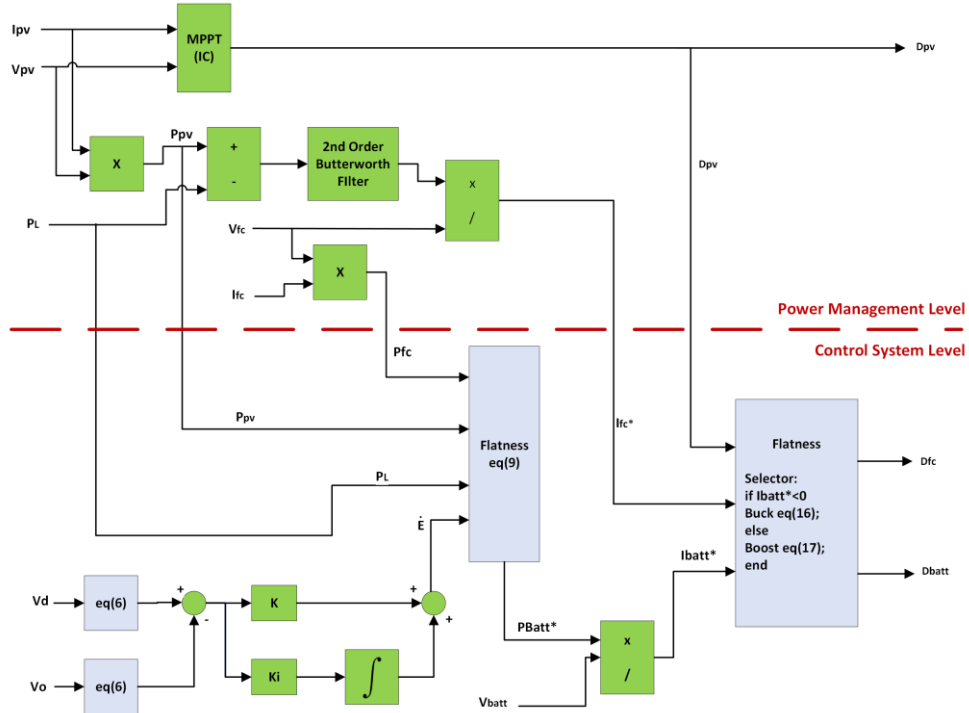


Figure 10. Proposed Power Management and Control System of Hybrid System

The overall structure of the control system remains similar to the double-flatness approach. However, the key difference lies in the direct application of the duty cycle to the converter. This modification allows for more independent operation of the MPPT algorithm while maintaining the benefits of the flatness-based control for the fuel cell and battery components.

Chapter 4 – Power Management Systems: State Flow and Linear Programming

The purpose and goals of PMS for higher level strategy were discussed in Chapter 2.3. As mentioned before, PMS can be divided into heuristic and non-heuristic methods. This chapter will look more closely at both types, expanding on the basic ideas discussed earlier.

In this project, two specific approaches are focused on: the heuristic method using state-flow strategies and the non-heuristic method employing Linear Programming (LP). These methods are chosen for their distinct characteristics. It will later be shown that state-flow is characterized by robustness, while LP is known for its optimality.

4.1. State Flow Method

The objective of state flow is to provide adaptability, enabling the system to handle various constraints effectively. In this project, the main constraints are battery level and the minimum running time of the fuel cell.

State flow strategies are particularly useful in managing complex systems with multiple operating modes and transitions. These strategies enable the implementation of rule-based decision-making processes that can quickly adapt to changing conditions in the power system as outlined by the theory of discrete event systems (DES) (Cassandras et al, 2008).

In the context of power management, the state flow approach can be understood as a DES where the system's behavior is modeled using a state machine. Different states represent various operating conditions or modes, such as battery charging or fuel cell operation. Transitions between these states are triggered by events or conditions, such as changes in battery level or power demand.

This approach is called a discrete event system because the events do not occur continuously as at the control system level used for voltage regulation. Instead, these events happen discretely and are handled at the higher level strategy for power management if solar panel and fuel cell. This distinction is crucial in understanding the role of the higher level strategy in relation to the lower-level battery power regulation.

By applying the state flow strategy, we can formally model the different operating states of the power system, define clear transition conditions, and ensure that all possible system states are accounted for. This approach enhances the robustness of the PMS, allowing it to handle various constraints effectively and adapt to changing conditions in real time.

4.1.a. Description of Desired Characteristic

States Description

- Photovoltaic
 - The main objective is to harvest the maximum energy from the sun.

- Normal operation: MPPT algorithm for maximum power production
- Non-MPPT state: used when P_{PV} exceeds the P_L , and the battery is already full.
- The control system level eliminates the need for separated day/night states. P_{PV} has been treated as disturbance in the control system level.
- Fuel Cells
 - Operate in emergency conditions (e.g. high power load, low battery SOC).
 - Limited by the dynamics of a second-order Butterworth filter.
 - Provide excess power for battery charging during cloudy days or at night when P_{PV} is insufficient. This is important to ensure the battery can supply enough power during transitions.
 - Minimum operation duration: T seconds (to prevent damage and extend lifetime)
- Battery
 - Always back up the power that cannot be supplied by P_{PV} or P_{FC} or absorbed the power oversupplied by P_{PV} or P_{FC} to achieve power balance with the load (P_L) while ensuring voltage regularity. This does not apply when the system is in a non-MPPT state.

Based on the description above, the system has several states defined by the three components as follows:

1. Photovoltaic (PV): MPPT, non-MPPT
2. Fuel Cells (FC): On, Off
3. Battery: On, Off, Charge

The possible combination of these three components defines the system's state as detailed below:

PV	FC	Battery	State
MPPT	Off	Off	Not Possible
MPPT	Off	On	P3
MPPT	On	Off	Not Possible
MPPT	On	On	P1
MPPT	On	Charge	P2
On Non-MPPT	Off	Off	Not Possible
On Non-MPPT	Off	On	Not Possible
On Non-MPPT	On	Off	P4
On Non-MPPT	On	On	Not Possible
On Non-MPPT	On	Charge	Not Possible

Table 5. Potential State Flow States

4.1.b. State Descriptions

● P1: PV MPPT, FC on, Battery Charge

This state can generally be implemented using equations (15) and (16). However, to charge the battery, it is necessary to account for the excess power produced by the fuel cell that will be

received by the battery. For this purpose, a new variable K (in Watts) is introduced, representing the power to charge the battery during steady state. The implementation scheme is as follows:

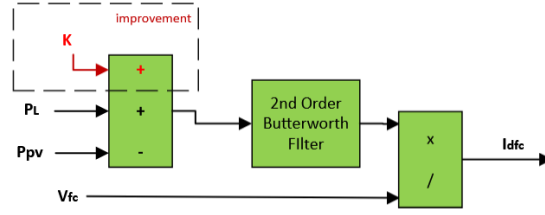


Figure 11. Implementation of Variable K

- **P2: PV MPPT, FC on, Battery on**

This state is similar to P1, except that $K=0$.

- **P3: PV MPPT, FC off, Battery on**

In this state, the same control strategy can be applied as in P1 or P2 but by setting $I_{dfc}=0$, as shown below:

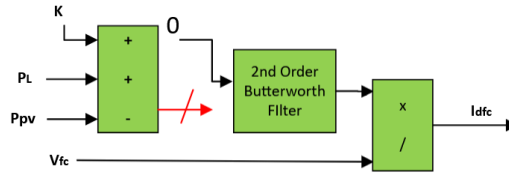


Figure 12. Scheme for P3 by Set Input for Filter to Zero

This scheme is more convenient for fuel cells, as it avoids an abrupt turning off the FC and cutting FC current to zero.

- **P4: PV on (Non-MPPT)**

In this case, equation (16) can be used by choosing $n = 2$, and setting I_{pv} as I_n , the flatness control can be applied again as bellow:

$$\alpha_{fc} = \frac{L_{fc}}{V_0} \left(\frac{V_0 - V_{fc}}{L_{fc}} - \lambda_{fc}(h_{fc} - h_{dfc}) + \dot{h}_{dfc} \right)$$

$$\alpha_{pv} = \left(\frac{V_{pv}V_{dc}}{L_{pv}} + \frac{2V_{dc}I_{batt}}{RC} \right)^{-1} \left(-\frac{V_{fc}(V_{fc} - V_{dc})}{L_{fc}} - \frac{V_{pv}(V_{pv} - V_{dc})}{L_{pv}} \right. \\ \left. + \frac{2V_{dc}}{R} \left(\frac{I_{fc} + I_{pv}}{C} - \frac{V_{dc}}{RC} \right) - \left(\frac{V_{fc}V_{dc}}{L_{fc}} + \frac{2V_{dc}I_{fc}}{RC} \right) \alpha_{fc} \right. \\ \left. - \lambda_{pv,0}(h_{pv} - h_{dpv}) - \lambda_{pv,1}(\dot{h}_{pv} - \dot{h}_{dpv}) + \ddot{h}_{dpv} \right) \quad (18)$$

Where $h_{pv} = \frac{1}{2}(L_{pv}i_{pv}^2 + L_{fc}i_{fc}^2 + CV_o^2)$ and $I_{dpv} = \frac{\dot{E} + P_L - P_{FC}}{v_{pv}}$ and $\dot{E}$ is obtained from linearizing feedback control.

4.1.c. Basic State Flow

After implementing the four states defined in the preceding section, the transitions need to be defined. Transitions are events or triggers that cause changes from one state to another. Once these transitions are defined, the proposed state flow diagram can be deduced as follows:

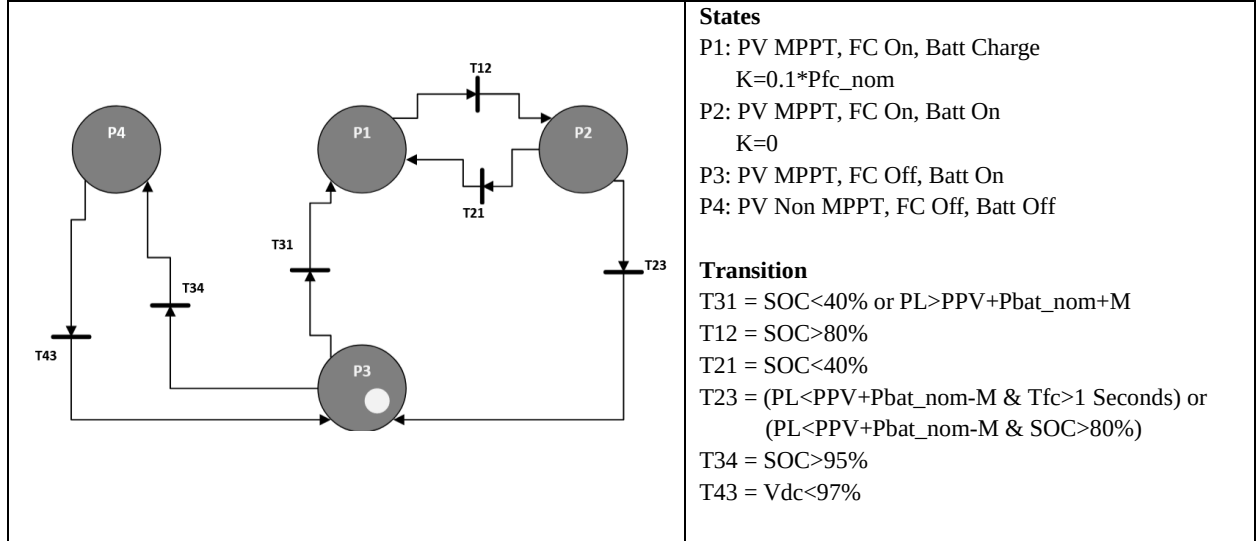


Figure 13. Basic State Flow

In MATLAB Simulink, there is a feature for State flow (MathWorks, 2024). It should be noted that the battery current cannot reach absolute zero due to internal resistance in electronics, which cannot be ideally modeled as zero in Matlab Simulink. Therefore, in steady state, the battery current is very small ($I_{bat} < 1A$), as shown below:

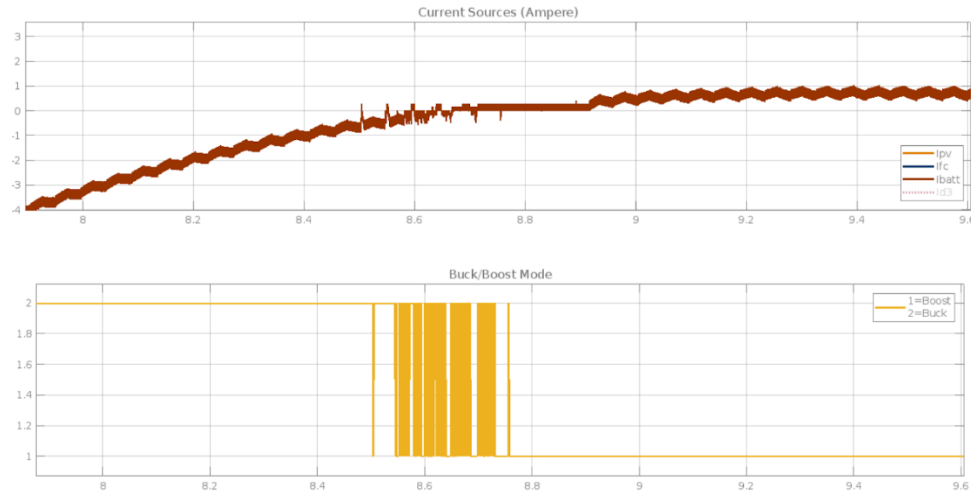


Figure 14. Steady State Condition in State No.2

Even though the battery current does not reach absolute zero and does not cause buck-boost oscillation, this steady-state condition should be avoided due to the presence of measurement noise in real situations, particularly from voltage and current of the PV system. Therefore, in the next improvement of the state flow design, the behavior in state no. 2 should be eliminated.

4.1.d. Improved State Flow

To address the issue of oscillation during steady state, an improvement in the state flow is proposed. The desired condition is that during fuel cell operation, the battery should not reach zero power but should instead remain slightly above or below the zero line. In this improved state-flow scheme, slow and fast states are introduced. The slow state is designed and lead to slow charging or discharging. Conversely, the fast condition is designed to accelerate the process of charging and discharging.

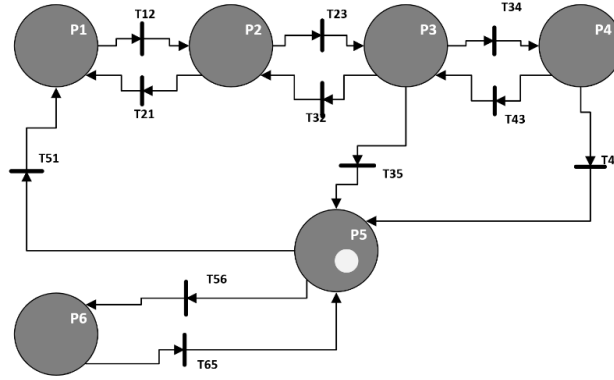


Figure 15. Improved State Flow

Places	Transitions	
P1-P4: PV MPPT, FC On, Batt On P1: $K=0.1 \cdot P_{fc_nom}$ (Fast Charge) P2: $K=0.0.5 \cdot P_{fc_nom}$ (Slow Charge) P3: $K=0.0.5 \cdot P_{fc_nom}$ (Slow Discharge) P4: $K=0.1 \cdot P_{fc_nom}$ (Fast Discharge) P5 PV MPPT, FC Off, Batt On P6: PV Non MPPT, FC Off, Batt Off	T12= SOC>40% T21= SOC<26% T23= SOC>60% T32= SOC<40% T34= SOC>73% T43= SOC>60%	T51= SOC<26% or $PL > P_{pv} + P_{bat_nom} + M$ T35= $PL < P_{pv} + P_{bat_nom} - M$ & $T_{fc} > T$ Seconds T45= $(PL < P_{pv} + P_{bat_nom} - M$ & $T_{fc} > T$ Seconds) or $(PL < P_{pv} + P_{bat_nom} - M$ & SOC>86%) T56= SOC>100% & $PL < P_{pv}$ T65= VDC<97%

Table 6. State and Transition of Improved State Flow

From table 6, this improvement relates to the variable K, which has four possible values. In the previous state flow scheme, K was designed to keep the battery in charging condition during steady state. In this improvement, values of K are varied, so depending of the value of K, the battery can be in one of the following conditions: Fast Charge, Slow Charge, Slow Discharge or Fast discharge.

Therefore, the State is P1 to P4 are developments from P1 in the previous state flow, and P5 and P6 correspond to P2 and P3 respectively, from the last state flow. Additionally, the transitions between states have been further developed as well, as detailed in table 6.

By setting the parameter values of the improved state flow, as shown in Table 7 below, in Matlab-Simulink:

Parameters	Value	Parameters	Value
0% Soc	7.99 A.hr	86% Soc	8.0075 A.hr
13% Soc	7.9925 A.hr	100% Soc	8.01 A.hr
26% Soc	7.995 A.hr	P_{fc_nom}	4000 W
40% Soc	7.9975 A.hr	P_{bat_nom}	2500 W
60% Soc	8.0025 A.hr	M (hysteresis)	100 W
73% Soc	8.005 A.hr	T (FC min work duration)	1 s

Table 7. State Flow Parameters for Matlab Simulation

The simulation result is obtained as shown below:

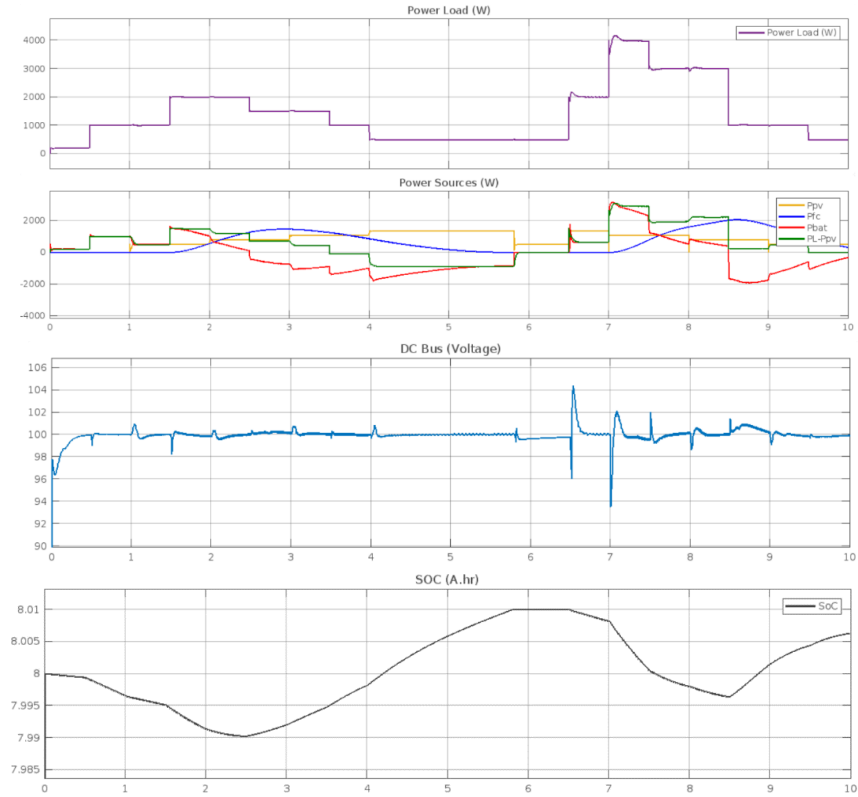


Figure 16. Matlab Simulation Result with Improved State Flow

From the simulation above, it can be observed that the hybrid system operating within the constraint region while maintaining the DC bus voltage around 100V, with an error of about 5%. With the status of states as follows:

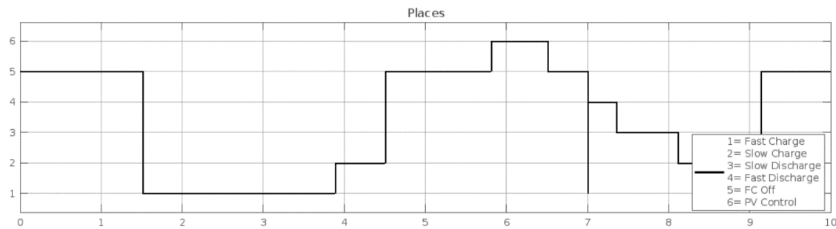


Figure 17. Status of States Over Time

From figure above, the hybrid power initially operates in state P5 (PV MPPT, FC Off, Batt On). This state is maintained until around 1.5 seconds when the battery SoC reaches 26%, causing a transition to state P1 (PV MPPT, FC off, Batt On: Fast Charge). At $t=3.8$ seconds, when the SoC exceeds 60%, and at the same time, the fuel cell operating duration meets the condition $t_{fc} > T$, leading the system to transition directly to P3 and then back to P5.

Subsequently, the load decreases significantly while solar power remains high, causing the battery SoC to reach 100% and triggering a state change to P6 at $t=5.7$ seconds. At $t=6.5$ seconds, the load increases again, surpassing the solar power output. This higher load causes the DC bus voltage to drop, triggering T65 and resulting in a return to state P5.

However, at $t=7$ seconds, the load increases to 4kW, necessitating the activation of the fuel cell despite the battery SoC remaining high. As a result, the system transitions to P1 and immediately shifts to P4 to enter fast discharging mode.

Therefore, it can be concluded that the state-flow is functioning very well. It transitions seamlessly from one state to another. In complete, power management and control system is established as follows:

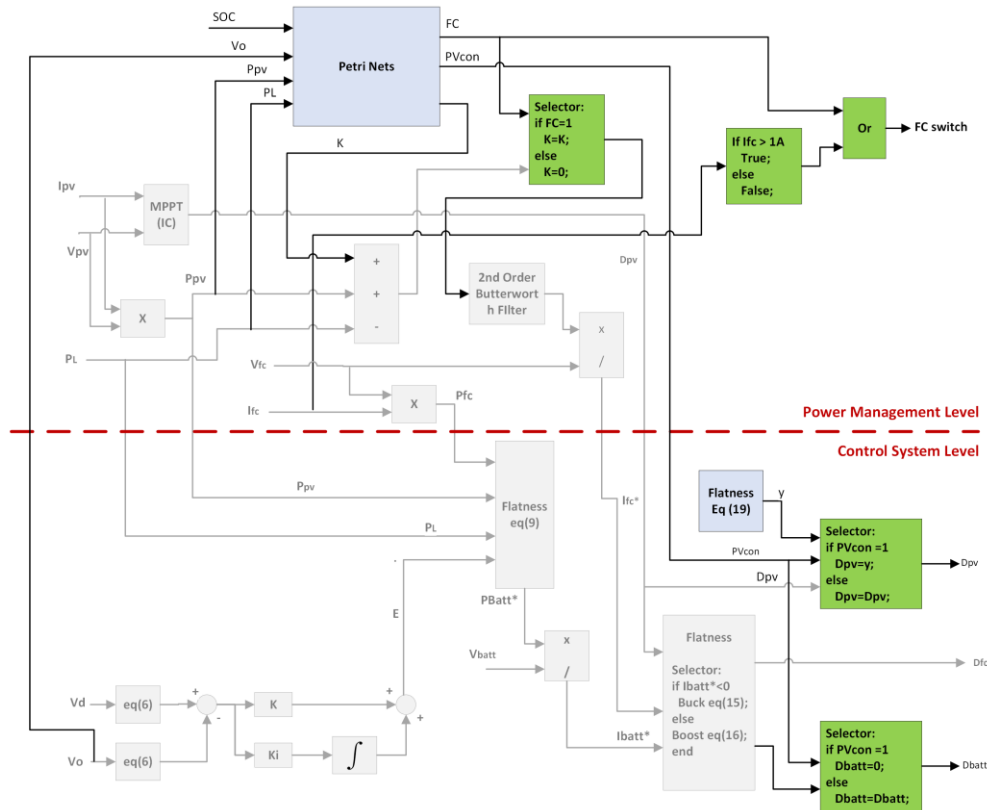


Figure 18. Heuristic Power Management and Control of Hybrid System

A quick note on the figure above, despite the complexity of the state flow scheme, it ultimately provides only three outputs. First, the FC command (discrete) is used to turn the fuel cell on and

off. Second, the PVcon command (discrete) disconnects the battery and uses the PV duty cycle to regulate the DC bus voltage. Finally, the K variable forces the battery into Fast/Sow Charge or Fast/Slow Discharge.

4.2. Linear Programming (LP) Method

In the non-heuristic method, mathematical and optimization techniques are used to solve the problem instead of a rule-based approach like the state flow method discussed in the preceding section. Therefore, the formulation of the problem, including the objective function and constraints, must be carefully examined (Bertsimas & Tsitsiklis, 1997).

The primary objective is to minimize the use of the fuel cell. To achieve this, the strategy involves maximizing the utilization of power from the battery, within the constrained of its capacity. Optimization techniques are used to plan the fuel cell's power output, ensuring that it meets the required demand at appropriate times. Consequently, the objective function focuses on minimizing the use of the fuel cell (Wang & Nehrir 2008).

Achieving this objective necessitates accurate predictions of future load and solar power, which can be accomplished using techniques such as artificial intelligence (Li et al., 2019). However, for the purposes of this project, predictions of power load and solar power are not included and are considered out of scope. The power demand, defined as $P_L - P_{PV}$, is obtained from the last state flow simulation to compare the performance of both heuristic and non-heuristic methods used in this project. Therefore, the power demand is assumed to be predicted with high accuracy for a specific interval.

From the description above, the objective function can be defined as follows:

$$f = \min \sum_{t=1}^T P_{fc} \quad (19)$$

By defining the power of demand as $P_d = P_{load} - P_{pv}$, the equality constraint is expressed as:

$$\begin{aligned} P_{fc}(t) + P_{batt}(t) &= P_d(t) \\ P_{fc}(1) &= 0 \end{aligned} \quad (20)$$

The inequality constraints are defined as:

$$SOC_{min} \leq SOC(t) \leq SOC_{max} \quad (21)$$

Where the relation between SOC and P_{batt} (in second) is given by:

$$SOC(t+1) = SOC(t) - \frac{P_{bat}}{3600} \quad (22)$$

Defining $x = (P_{bat}(1), \dots, P_{bat}(T), P_{fc}(1), \dots, P_{fc}(T), Soc(1), \dots, soc(T+1))^T$ and $T =$ Horizon Time shows that this is a linear problem:

$$\begin{aligned}
& \min_x f^t x \\
& \text{s.t.} \quad A \cdot x \leq b \\
& \quad \quad A_{eq} \cdot x = b_{eq} \\
& \quad \quad lb \leq x \leq ub
\end{aligned}$$

4.2.a. Linear Programming Simulation

Using MATLAB's linprog function, which implements various algorithms for solving linear programming problems (Mathworks, 2023b) and the power demand (P_{load} - P_{pv}) obtained from the last state-flow simulation, the power plan for the fuel cell is determined as follows:

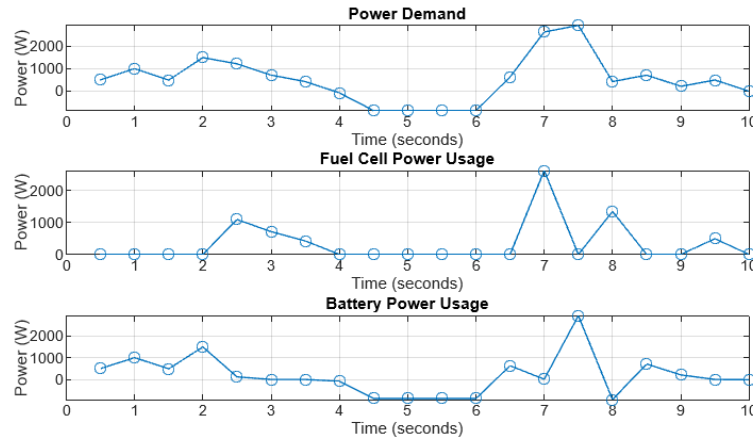


Figure 19. The Fuel Cell Power Trajectory Planned by LP

By setting the fuel cell power output as desired power for the PMS, a simulation is conducted as shown below. However, since the LP plan does not account the limitation of fuel cell dynamics, the trajectory is 1.5s faster than planned in LP.

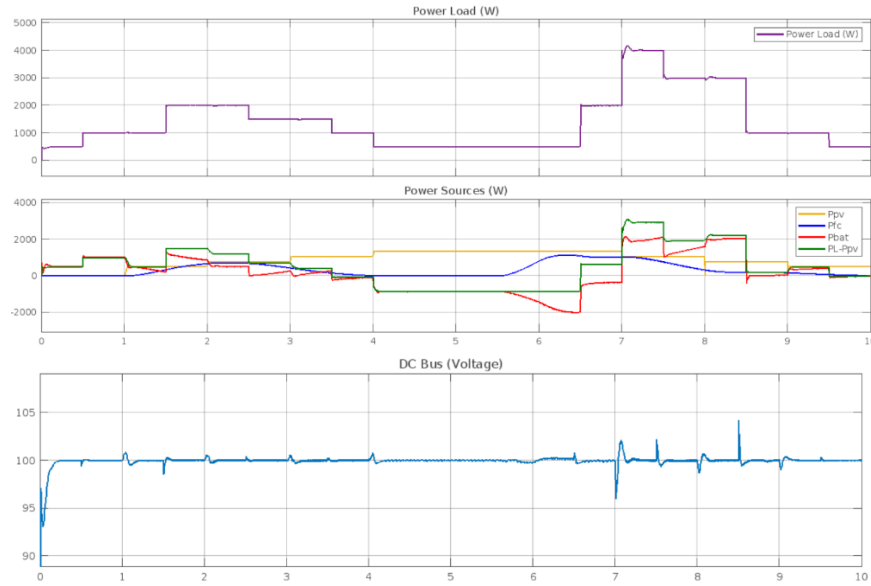


Figure 20. The Simulation Result after Imposing the Fuel Cell Trajectory

The SOC comparison:

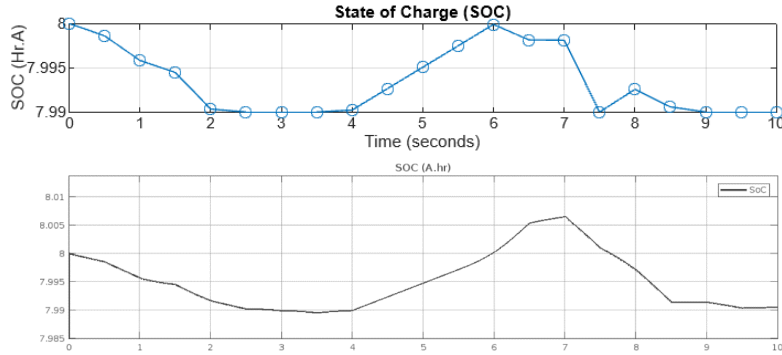


Figure 21. The SOC Graphic of LP Plan (Above) and Simulation Result (Below)

From Figure 20, it can be observed that the trajectory of SOC is not similar to the SOC trajectory plan. This discrepancy is caused by the fuel cell dynamic limitations not being included in the linear programming formulation. In the next section, the fuel cell dynamic limitation will be added to the LP constraints, and the effect of the Butterworth filter will be reduced to improve the accuracy of the model.

4.2.b. Linear Programming Improvement

In this section, to achieve the same trajectory as the plan, an additional constraint in the LP formulation is required. By incorporating a constraint on the maximum fuel cell power increase, the dynamic behavior of the fuel cell can be limited.

$$\begin{aligned} P_{fc}(t) - P_{fc}(t + 1) &< \text{Max_FC_Increase} \\ P_{fc}(t) - P_{fc}(t + 1) &> -\text{Max_FC_Increase} \end{aligned} \quad (23)$$

However, since smooth dynamic performance of the fuel cell is still necessary, the Butterworth filter should not be completely removed. Instead, its effect can be reduced by increasing its order.

After increasing the order of the filter and simultaneously limiting the power increase of the fuel cells to 1000 W/s, the LP produces the following trajectory plan:

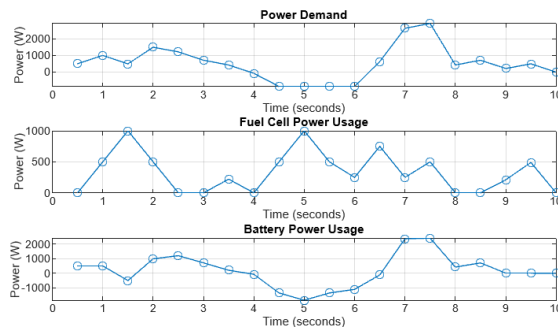


Figure 22. The Fuel Cells Power Trajectory Planned by Modified LP

The simulation results are shown below:

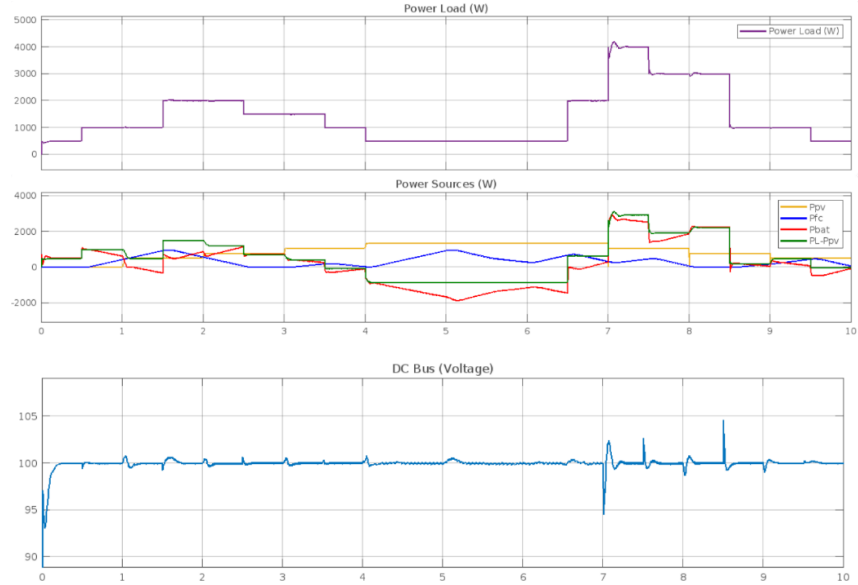


Figure 23. The Simulation Result after Imposing the Improved Fuel Cell Trajectory

The comparison of SOC and power accumulation is shown below. It can be observed that the LP plan and the simulation results exhibit mostly similar graphic after the modifications.

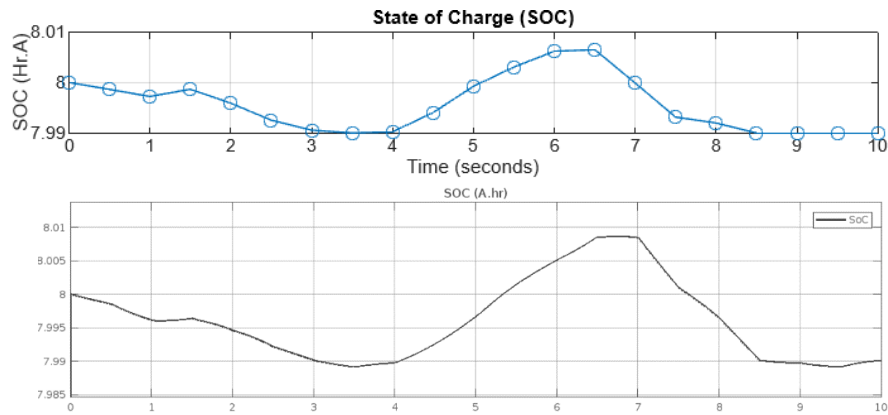


Figure 24. The SOC Graph of Improved LP Plan (Above) and Simulation Result (Below)

From figure 23, it is evident that although the SOC result is not 100% identical to the plan, it is closely similar. The same observation is made for the energy trajectory, where the result is quite similar to the plan, as shown in the figure below:

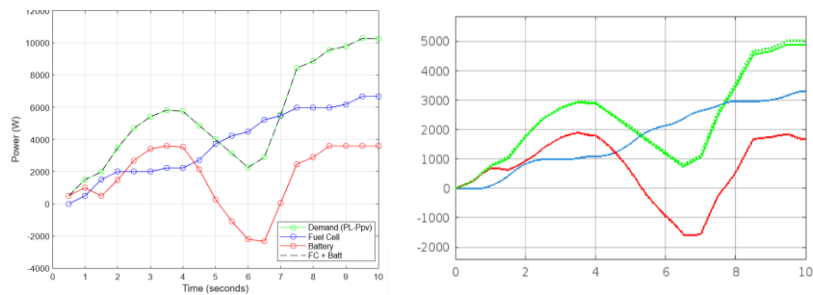


Figure 25. The Energy (J) Plot of Improved LP (Left) and Simulation Result (Right)

Therefore, it can be concluded that the modified LP plan is reliable and can be used as the PMS for the hybrid system. Hence, the complete power management and control system with the non-heuristic method is shown below:

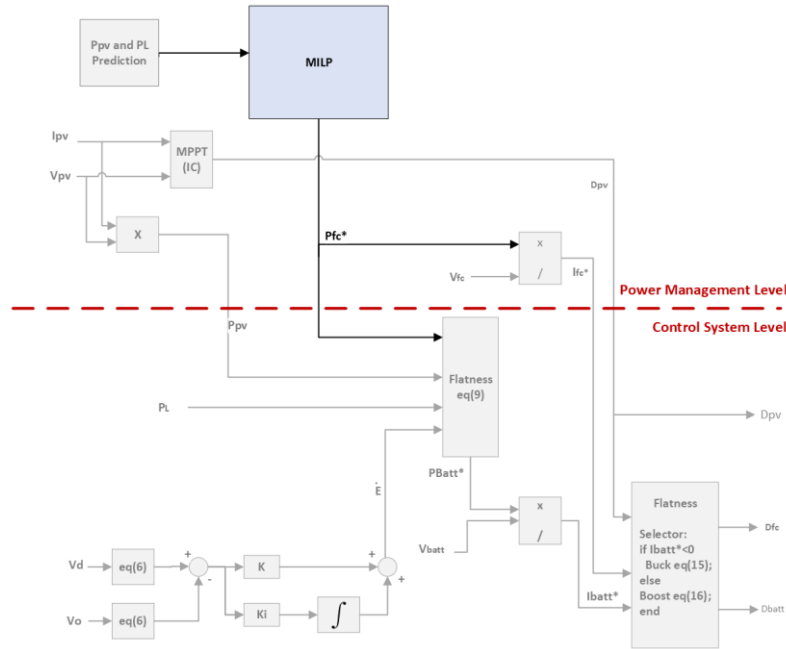


Figure 26. Power Management and Control System of Hybrid System with Linear Programming

4.2.c Availability of Linear Programming Solution

In this section, the robustness of the LP method will be evaluated by introducing uncertainties to the power demand. Specifically, white noise ranging from 0-200W, varying every 0.5s around the initial power demand, will be added.

The effect of horizon selection on the LP implementation will also be assessed. Previously, the horizon was fixed at 10 seconds during the simulation, as was the power demand. This section will investigate whether varying the horizon affects the LP's performance. Since the load in the previous case lasted only 10 seconds, it is assumed that the same power demand pattern continues for the next 10s, allowing for the application of a 10-second horizon.

By running the simulation 100 times, the success rate of LP in providing solution is shown in the table below:

Horizon (s)	White noise 200 W	White noise 200W with a 1000 W noise at t=5 s
2.5	79%	49%
10	82%	81%

Table 8. Success Rate of LP in Providing Solution

One example of a fuel cell trajectory with noise can be seen in Figure 26. It demonstrates that, in the end (at $t = 10s$), the difference between the actual power accumulation and the initial power demand trajectory is 500 W, which results from a spike noise lasting for 0.5s at $t=5s$.

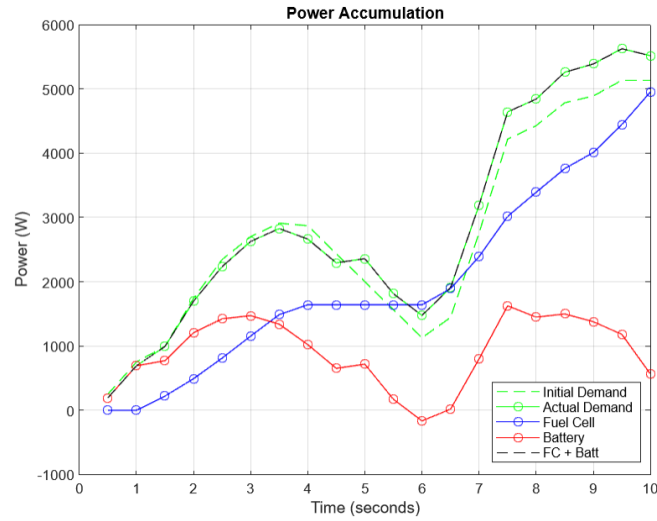


Figure 27. Energy (J) Plot for LP Trajectory with Power Demand Uncertainty

From the table above, it can be observed that LP can solve the problem even with sudden changes to some extent. Additionally, it is evident that a longer horizon generally increases the likelihood of finding a solution with LP.

Chapter 5 – Result and Analysis

In this chapter, the simulation results from both the state flow and LP methods will be discussed. By comparing these two methods, it can be concluded that the LP method is more efficient. The results are shown in the table and figure below:

Source	State Flow (J)	LP (J)
Fuel cell	6630	3331
Battery	-1185	1685
Net Energy consumed	5445	5017

Table 9. Energy Flow of Fuel Cell and Battery

From the table above, it can be observed that the state flow method consumes nearly twice as much fuel cell energy as the LP method. However, the state flow method results in a higher amount of energy stored in the battery at the end, while in contrast, the LP method consumes the energy initially stored in the battery. This highlights the different characteristics of the two methods: the state flow method demonstrates a robust strategy, while the LP method optimizes energy usage by efficiently utilizing the energy stored in the battery at each moment.

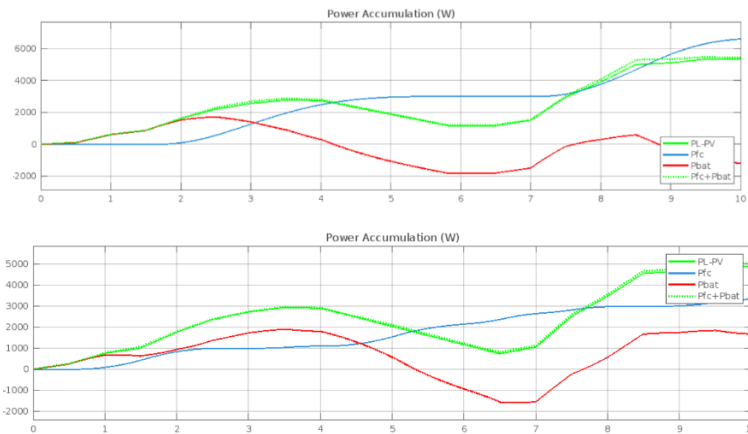


Figure 28. Energy Plot: State Flow (Above) and Linear Programming Method (Below)

A particularly interesting observation is in the net energy consumption. The data demonstrates that for the same load, the LP method consumes less energy than the state flow method, specifically saving 428J of energy. In the state flow method, the PMS does not fully utilize the maximum power from the solar panels. As described earlier, when the SOC is fully charged and the power load is lower than the maximum solar power, state of number 6 occurs, which consequently does not utilize the maximum solar power. This situation was observed during the interval from approximately 5.8s to 6.5s in the state flow simulation (Chapter 4.1.4).

In contrast, the LP method allows for the anticipation of such states because the optimization is calculated based on a defined horizon time. By setting the SOC limits between 0% and 100%, the LP method continuously aims to avoid these limits while simultaneously optimizing the use of fuel cell and battery energy.

However, the LP method may face challenges when it fails to find an optimal solution as discussed in chapter 4.2c. While the LP-based method allows for better planning of power management to maximize solar power absorption, it is heavily reliant on the accuracy of power load and solar power predictions. Inaccurate predictions can lead to a non-solution condition where the battery exceeds its limits. Even with accurate predictions, a non-solution condition can occur if solar energy is abundant and the battery capacity is insufficient to absorb all the energy. Inaccurate predictions can exacerbate this problem.

Therefore, there is a tradeoff between robustness and efficiency between the two methods, primarily due to State 6. In State 6, the system may lose some solar energy, but this state also plays a crucial role when solar energy is abundant and the battery cannot absorb all the energy (robustness).

Chapter 6 - Conclusion and Future Work

In this project, two methods of power management and control systems for regulate power of battery (lower level) and two methods for regulate power of solar panel and fuel cell (higher level) were discussed.

For the lower level, control with flatness-PID and double flatness was conducted. From the simulation, it was concluded that there is no significant performance difference between these two methods. The double flatness design has not shown any improvement in performance compare to flatness-PID control. This is because the systems operate in a linear region (Fliess et al., 1994) after applying the first flat output (9). As a result, once the system enters the linear region, the second flatness equations (13), (14) do not offer a significant performance advantage over classical methods such as PID controls.

However, the use of second flatness is intriguing from the perspective electrical modelling and control theory. Further investigation is recommended to explore how second flatness (13), (14) can be utilized effectively for converter control without the first flatness (9).

Second, two method of higher level which are state flow method and Linear Programming method have been discussed. The advantages and disadvantages between these two methods are summarized as follows:

Method	Advantage	Disadvantage
State Flow Method	• More robust • In case of excess solar power, State 6 can take place, by performing PV control	• Potential loss of solar power utilization, particularly in State 6. •
LP Method	• Provide optimum solution	• Even in the high accuracy power demand, the LP may lead to non-solution problem. This only happen when excess power cannot be absorbed by a full SOC battery. In heuristic method this case can be anticipated by State 6. • Relies heavily on prediction accuracy. Poor accuracy increases the likelihood of a non-solution problem.

Table 10. Advantage and Disadvantage of State flow and LP Methods

To achieve both optimality and robustness, future work should focus on the following areas:

1. Improving prediction methods: Integrating artificial intelligence for weather condition prediction could significantly enhance the accuracy of power generation forecasts,

particularly for solar energy. This improved accuracy would benefit the LP method, reducing instances of non-solution problems.

2. Optimal component sizing: Investigating the optimal sizing of components and equipment, such as batteries. This ensures that the system's optimality is always guaranteed without oversizing. Proper battery sizing will help energy storage and utilization more effectively, potentially mitigating issues in both the LP and state flow methods. A recent study (Khawaja et al. 2023) proposed an integrated framework for sizing and energy management of hybrid energy systems using finite automata, which considers both technical and economic factors in optimizing system components.
3. Integration of prediction with state flow methods: While this project utilized prediction in the non-heuristic (LP) method, integrating prediction capabilities with the state flow method would enhance its performance. This integration could provide a more robust system that can anticipate and adapt to changing conditions. This idea was suggested by the supervisor, but due to a tight deadline, this development could be pursued in future projects. However, the non-heuristic such as LP was chosen initially because it is intuitive to optimize when predictions are known, especially with high accuracy.
4. Hybrid approach: Combining both state flow and LP methods could leverage the strengths of each approach. For example, when the LP method cannot pose any optimal solution, the state flow method could take over until the LP method able to find optimal solution again.
5. Real-time implementation: Future efforts should aim to implement power management in real-time/online, covering both optimization and prediction. In this project, the LP approach works offline and does not include a prediction step. The LP solution was applied manually after the LP plan trajectory was executed in a separate m-file. Implementing Model Predictive Control for PMS could be explored to address this limitation.
6. Hardware-In-the-Loop (HIL) testing: Conducting HIL tests, for example using Simulink Real-time with Speedgoat FPGA Hardware (Mathworks, 2023c), would provide valuable insights into the real-world performance and robustness of the developed control and power management systems.

These future work aim to address the limitations identified in this project and improve the more efficient, robust, and adaptable PMS for renewable applications.

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Appendices

Appendix 1. MPPT Algorithm

```
function y = MYMPP(u,i,uo,io,D)

du=u-u0;
di=i-io;
d=0.001;
if du==0
    if di==0
        m=D;
    else
        if di>0
            m=D-d;
        else
            m=D+d;
        end
    end
else
    if di/du== -(i/u)
        m=D;
    else
        if di/du> -(i/u)
            m=D-d;
        else
            m=D+d;
        end
    end
end
y = m;
end
```

Appendix 2. Average Model

Averaging Method

Theoretically (Hassan et al., 1996), the averaging method applies to a system with the form:

$$\dot{x} = \varepsilon f(t, x, \varepsilon)$$

Where ε is a small positive parameter and $\varepsilon f(t, x, \varepsilon)$ is T-periodic in time, meaning:

$$f(t + T, x, \varepsilon) = f(t, x, \varepsilon), \quad \forall (t, x, \varepsilon) \in [0, \infty) \times D \times [0, \varepsilon_0]$$

For some domain $D \subset R^n$. The method approximates the solution of the system by the solution of an averaged system, obtained by averaging $f(t, x, \varepsilon)$ at $\varepsilon = 0$.

For the DC/DC converter which works in two modes, the averaging method can be simplified as follows:

$$f_{av}(x) = \frac{1}{T} \int_0^T f(\tau, x, 0) d\tau = d \times f_1(\tau, x, 0) + (1 - d) \times f_2(\tau, x, 0)$$

Where $f_1(\tau, x, \varepsilon)$ and $f_2(\tau, x, \varepsilon)$ refer to state equations when switch is closed and opened, respectively, and d is duty cycle of the pulse. Where the close and open state are described as follows:

$$\dot{x} = A_1x + B_1u$$

$$\dot{x} = A_2x + B_2u$$

Then the average state space model is:

$$\dot{x} = \bar{A}x + \bar{B}u$$

Where $\bar{A} = A_1d + A_2(1 - d)$ and $\bar{B} = B_1d + B_2(1 - d)$.

A simple proof is given below (Philip et al., 1998). Assume the inductor current and capacitor voltage change linearly in each working mode. Then, in one T period, the following equations hold:

$$x(dT) = x(0) + \dot{x} \times dT = x(0) + (A_1x + B_1u) \times dT$$

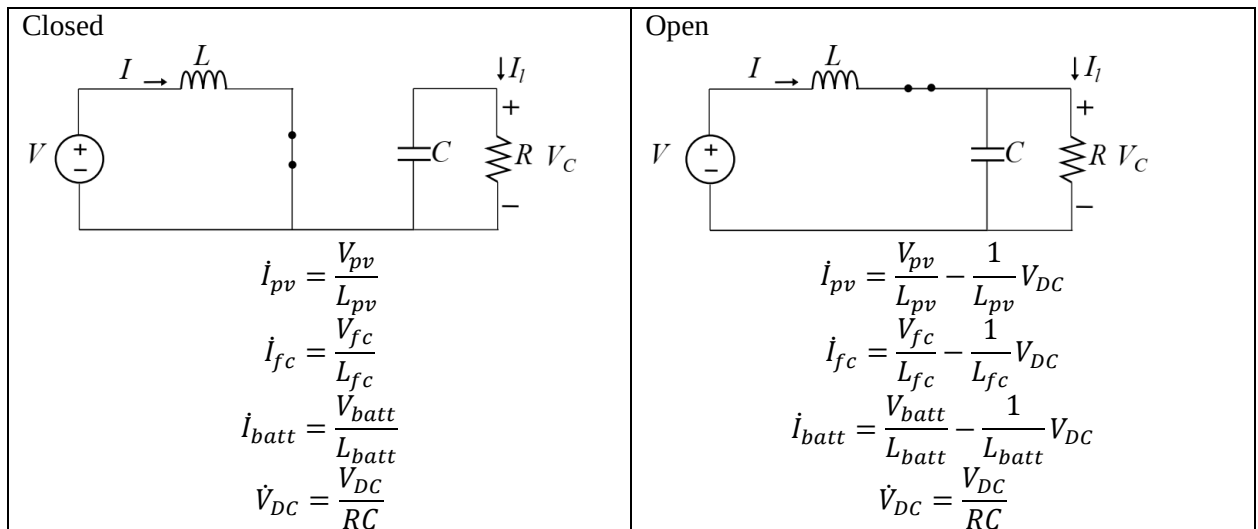
$$\begin{aligned} x(T) &= x(dT) + \dot{x} \times (1 - d)T = x(dT) + (A_2x + B_2u) \times (1 - d)T \\ &= x(0) + (A_1x + B_1u) \times dT + (A_2x + B_2u) \times (1 - d)T \\ &= x(0) + [(A_1d + A_2(1 - d))x + (B_1d + B_2(1 - d))u]T \end{aligned}$$

Therefore,

$$\dot{x} = \frac{x(T) - x(0)}{T} = (A_1d + A_2(1 - d))x + (B_1d + B_2(1 - d))u = \bar{A}x + \bar{B}u$$

Boost Converter Average model

From the converter's Simulink scheme in figure 5, the closed and open states of the boost converter are as follows:

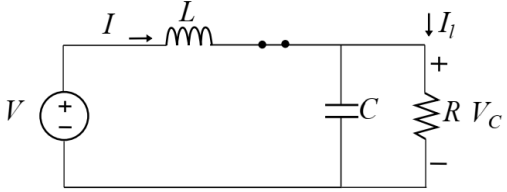
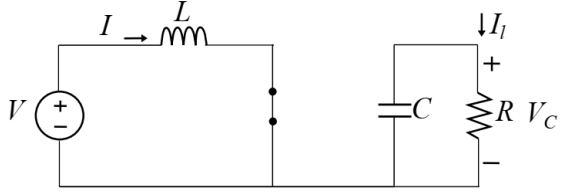


The state equations for the closed and open are written as follows:

Closed $\begin{bmatrix} \dot{i}_{pv} \\ \dot{i}_{fc} \\ \dot{i}_{batt} \\ \dot{V}_{DC} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{-1}{RC} \\ 0 & 0 & 0 & \frac{-1}{RC} \end{bmatrix} \begin{bmatrix} I_{pv} \\ I_{fc} \\ I_{batt} \\ V_{DC} \end{bmatrix} + \begin{bmatrix} \frac{V_{pv}}{L_{pv}} \\ \frac{V_{fc}}{L_{fc}} \\ \frac{V_{batt}}{L_{batt}} \\ 0 \end{bmatrix}$	Open $\begin{bmatrix} \dot{i}_{pv} \\ \dot{i}_{fc} \\ \dot{i}_{batt} \\ \dot{V}_{DC} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \frac{-1}{L_{pv}} \\ 0 & 0 & 0 & \frac{-1}{L_{pv}} \\ 0 & 0 & 0 & \frac{-1}{L_{batt}} \\ \frac{1}{C} & \frac{1}{C} & \frac{1}{C} & \frac{-1}{RC} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ V_o \end{bmatrix} + \begin{bmatrix} \frac{V_{pv}}{L_{pv}} \\ \frac{V_{fc}}{L_{fc}} \\ \frac{V_{batt}}{L_{batt}} \\ 0 \end{bmatrix}$
---	---

Using the average model equation above, we can obtain equation (12)

Buck converter Average model

Close  $\dot{i}_{batt} = \frac{V_{batt}}{L_{batt}} - \frac{1}{L_{batt}} V_{DC}$ $\dot{V}_{DC} = \frac{V_{DC}}{RC}$	Open  $\dot{i}_{batt} = \frac{V_{batt}}{L_{batt}}$ $\dot{V}_{DC} = \frac{V_{DC}}{RC}$
--	--

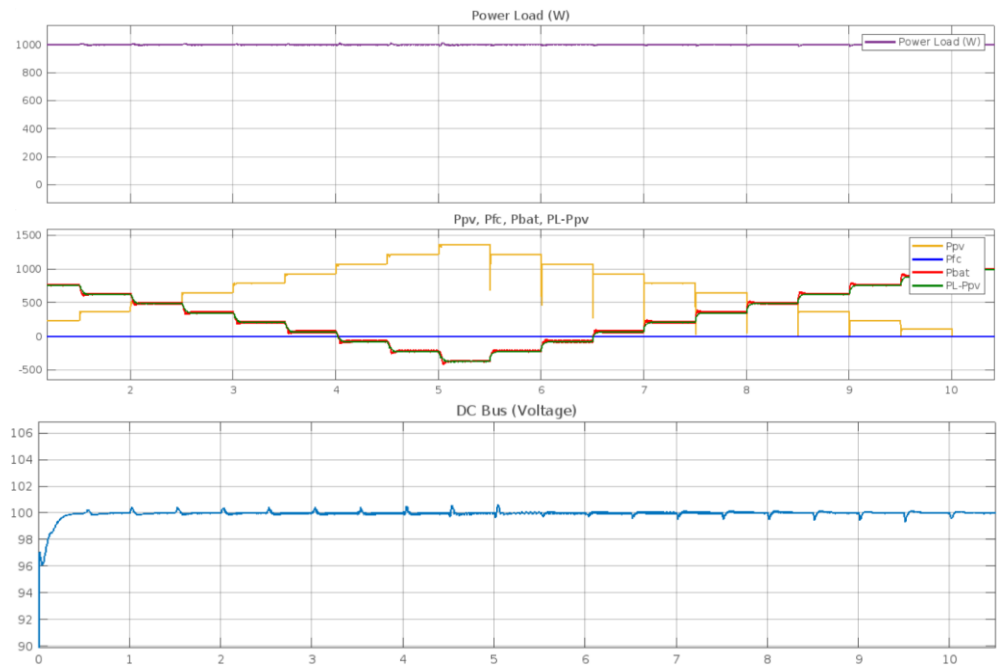
Which written in state equation is as bellow

Close $\begin{bmatrix} \dot{i} \\ \dot{V}_C \end{bmatrix} = \begin{bmatrix} 0 & \frac{-1}{L_{batt}} \\ \frac{1}{C} & \frac{-1}{RC} \end{bmatrix} \begin{bmatrix} I_{batt} \\ V_{DC} \end{bmatrix} + \begin{bmatrix} \frac{V_{batt}}{L_{batt}} \\ 0 \end{bmatrix}$	Open $\begin{bmatrix} \dot{i}_{batt} \\ \dot{V}_{DC} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & \frac{-1}{RC} \end{bmatrix} \begin{bmatrix} I_{batt} \\ V_{DC} \end{bmatrix} + \begin{bmatrix} \frac{V_{batt}}{L_{batt}} \\ 0 \end{bmatrix}$
--	---

Using average model equation above we can obtain equation (11).

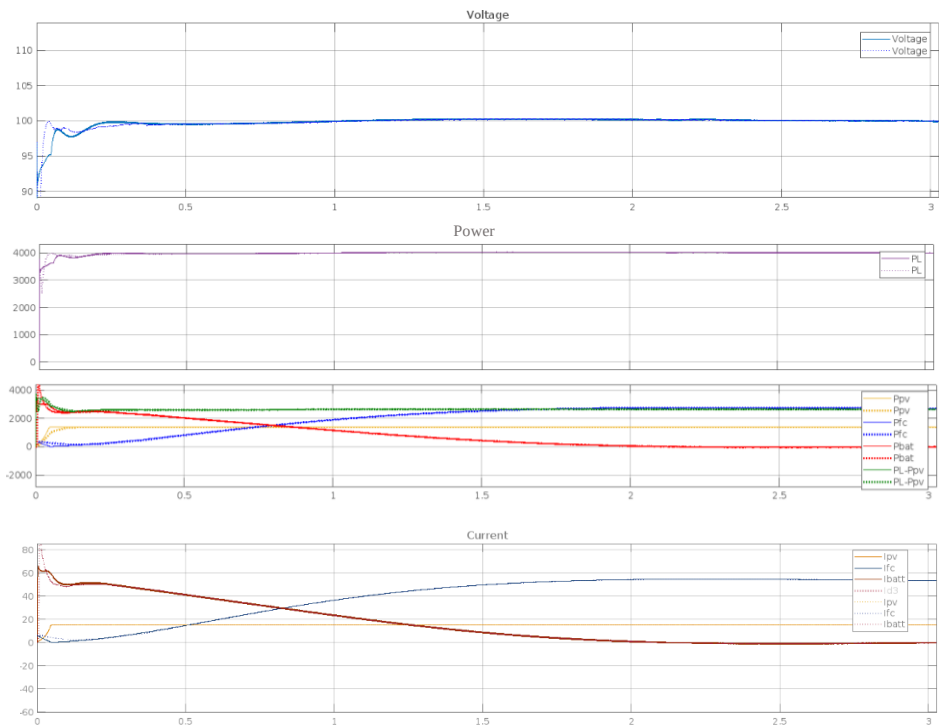
Appendix 3. Model Validation

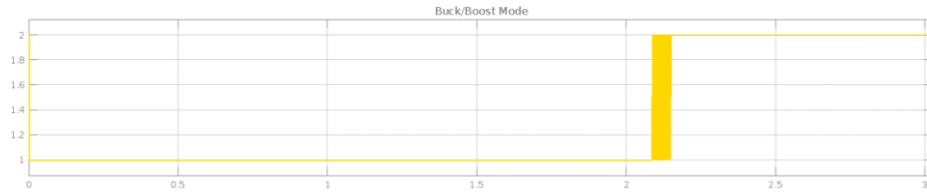
MPPT Validation



Voltage (V) and Power (W) Graph with Various Irradiance (W/m²)

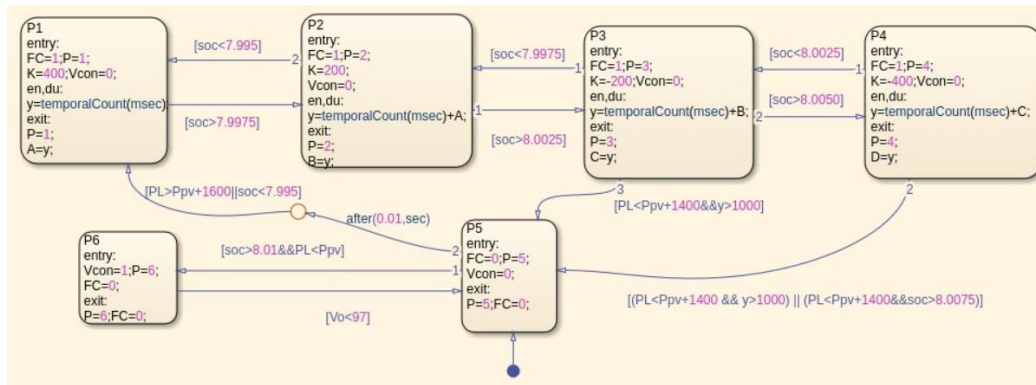
Control Validation during Transient





Comparison of Two Control Method during Transient

Appendix 4. Matlab Simulink State Flow



State Flow Design in Matlab Simulink

Appendix 5. LP Fuel Cell Trajectory Planning Code

```
% Define parameters
total_time_horizon = 20; % total simulation time
prediction_horizon = 20; % MPC prediction horizon
initial_SOC = 8; % Initial SOC in Hr.A
SOC_min = 7.99; % SOC minimum in Hr.A
SOC_max = 8.01; % SOC maximum in Hr.A
battery_capacity = 11.6; % Battery capacity in Hr.A
max_charge_power = 4000; % Maximum charge power in W
max_discharge_power = 4000; % Maximum discharge power in W
max_fuel_cell_power = 4000; % Maximum fuel cell power in W
max_fuel_cell_increase = 500; % Maximum increase in fuel cell power per 1/20th of time_horizon

% Initial power demand pattern in watts
P_demand_pattern = [500,1000,489.6, 1490,1210,...
    710, 421, -79,-871,-871,-871,...
    -871, 629, 2629, 2921,421, 710,210,490,-10];

% Initialize variables
SOC_initial = initial_SOC;
P_fuel_cell_result = zeros(1, total_time_horizon);
P_battery_result = zeros(1, total_time_horizon);
```

```

SOC_result = zeros(1, total_time_horizon + 1);
SOC_result(1) = initial_SOC;
actual_power_demand = zeros(1, total_time_horizon);
FC_initial=0;

% Real-time power demand update function (example function)
function updated_demand = getRealTimeDemand(t)
    % Here, we simulate a changing power demand pattern. In practice,
    % this function would get data from a real-time data source.
    base_demand = [500,1000,489.6, 1490,1210,...
    710, 421, -79,-871,-871,-871,...
    -871, 629, 2629, 2921,421, 710,210,490,-10];
    noise = 200*randn(size(base_demand)); % Random noise
    updated_demand = base_demand + noise;
    updated_demand = circshift(updated_demand, -mod(t, length(base_demand))); % Shift the pattern
end

for t = 1:total_time_horizon
    %disp(FC_initial)
    % Get the current prediction horizon demand pattern
    P_demand = getRealTimeDemand(t-1);
    %disp(P_demand)

    % Store the actual power demand at this time step
    actual_power_demand(t) = P_demand(1);
    actual_power_demand=actual_power_demand(1:20)

    % Define current prediction horizon
    horizon = prediction_horizon;

    %horizon = min(prediction_horizon, total_time_horizon - t + 1);

    % Define the optimization problem for the current horizon
    n = horizon * 2 + (horizon + 1); % P_battery, P_fuel_cell, SOC
    f = [zeros(1, horizon), ones(1, horizon), zeros(1, horizon + 1)];

    Aeq = zeros(horizon * 2, n);
    beq = zeros(horizon * 2, 1);

    for k = 1:horizon
        Aeq(k, k) = 1; % P_battery
        Aeq(k, horizon + k) = 1; % P_fuel_cell
        beq(k) = P_demand(k);

        if k < horizon
            Aeq(horizon + k, horizon * 2 + k) = 1; % SOC(k)
            Aeq(horizon + k, horizon * 2 + k + 1) = -1; % SOC(k+1)
        end
    end
end

```

```

    Aeq(horizon + k, k) = -1 / 100 / 3600; % P_battery(k) in W
end
end

Aeq_initial_soc = zeros(1, n);
Aeq_initial_soc(horizon * 2 + 1) = 1; % SOC(1)
beq_initial_soc = SOC_initial;

% Initial fuel cell power constraint (only at the first step)
Aeq_initial_fc = zeros(1, n);
Aeq_initial_fc(horizon + 1) = 1; % P_fuel_cell(1)
beq_initial_fc = FC_initial; % Initial fuel cell power is zero

Aeq = [Aeq; Aeq_initial_soc; Aeq_initial_fc];
beq = [beq; beq_initial_soc; beq_initial_fc];

A = zeros(horizon * 4, n);
b = zeros(horizon * 4, 1);

for k = 1:horizon + 1
    A(k, horizon * 2 + k) = 1; % SOC(k)
    b(k) = SOC_max; % Upper limit

    A(horizon + k, horizon * 2 + k) = -1; % SOC(k)
    b(horizon + k) = -SOC_min; % Lower limit
end

%Fuel cell power increase limits
for k = 1:horizon-1
    A(2 * horizon + k, horizon + k) = 1; % P_fuel_cell(t)
    A(2 * horizon + k, horizon + k + 1) = -1; % P_fuel_cell(t+1)
    b(2 * horizon + k) = max_fuel_cell_increase; % Maximum increase

    A(3 * horizon + k, horizon + k) = -1; % -P_fuel_cell(t)
    A(3 * horizon + k, horizon + k + 1) = 1; % P_fuel_cell(t+1)
    b(3 * horizon + k) = max_fuel_cell_increase; % Maximum decrease (symmetric)
end

% Additional constraint: P_fuel_cell(1) = 0
A(4 * time_horizon + 1, time_horizon + 1) = 1; % P_fuel_cell(1)
b(4 * time_horizon + 1) = 0; % Power of fuel cell at t=0 should be zero

% Lower bounds (P_fuel_cell >= 0, SOC constraints are already handled)
lb = [-max_discharge_power * ones(1, horizon), zeros(1, horizon), SOC_min * ones(1, horizon + 1)];

```

```
ub = [max_charge_power * ones(1, horizon), max_fuel_cell_power * ones(1, horizon), SOC_max  
* ones(1, horizon + 1)];
```

```
% Solve the optimization problem
```

```
[x, fval, exitflag] = linprog(f, A, b, Aeq, beq, lb, ub);
```

```
if exitflag == 1
```

```
    P_battery = x(1:horizon);
```

```
    P_fuel_cell = x(horizon + 1:2 * horizon);
```

```
    SOC = x(2 * horizon + 2); % SOC at next step
```

```
    SOC_initial = SOC;
```

```
    if t==1 || t== total_time_horizon
```

```
        FC_initial = P_fuel_cell(1);
```

```
    else
```

```
        FC_initial = P_fuel_cell(2);
```

```
    end
```

```
    % Store the results for the first control action
```

```
    P_battery_result(t) = P_battery(1);
```

```
    P_fuel_cell_result(t) = P_fuel_cell(1);
```

```
    SOC_result(t+1) = SOC;
```

```
else
```

```
    disp('Optimization did not converge.');
```

```
    break;
```

```
end
```

```
end
```

```
% Plot the results
```

```
time = 0.5:0.5:total_time_horizon / 2;
```